

Modern Technology

Sequencer Introduction

Sequencer Quickstart Guide

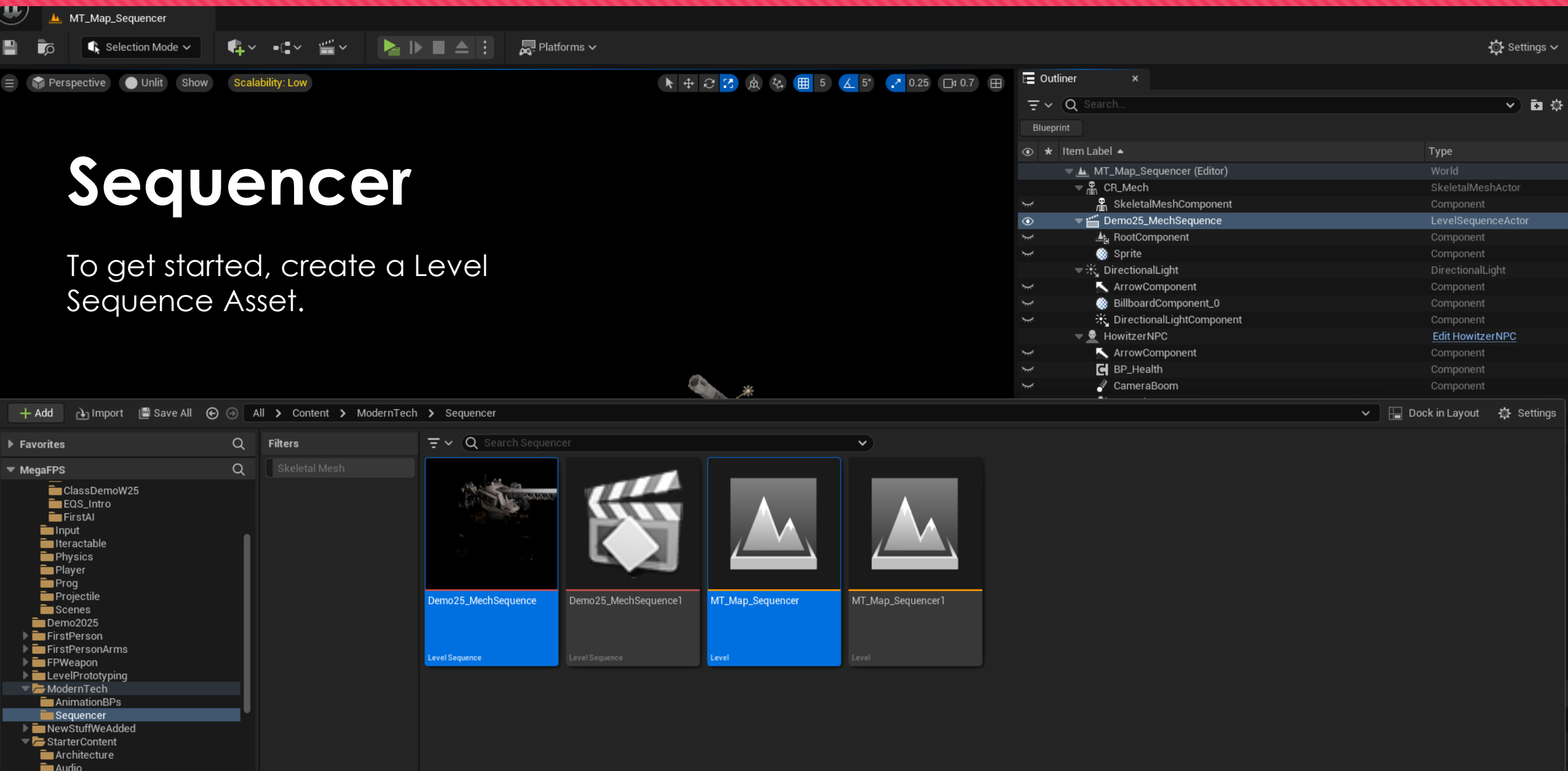
- Specific Sequencer Editor Docs:
https://dev.epicgames.com/documentation/en-us/unreal-engine/sequencer-cinematic-editor-unreal-engine?application_version=5.7

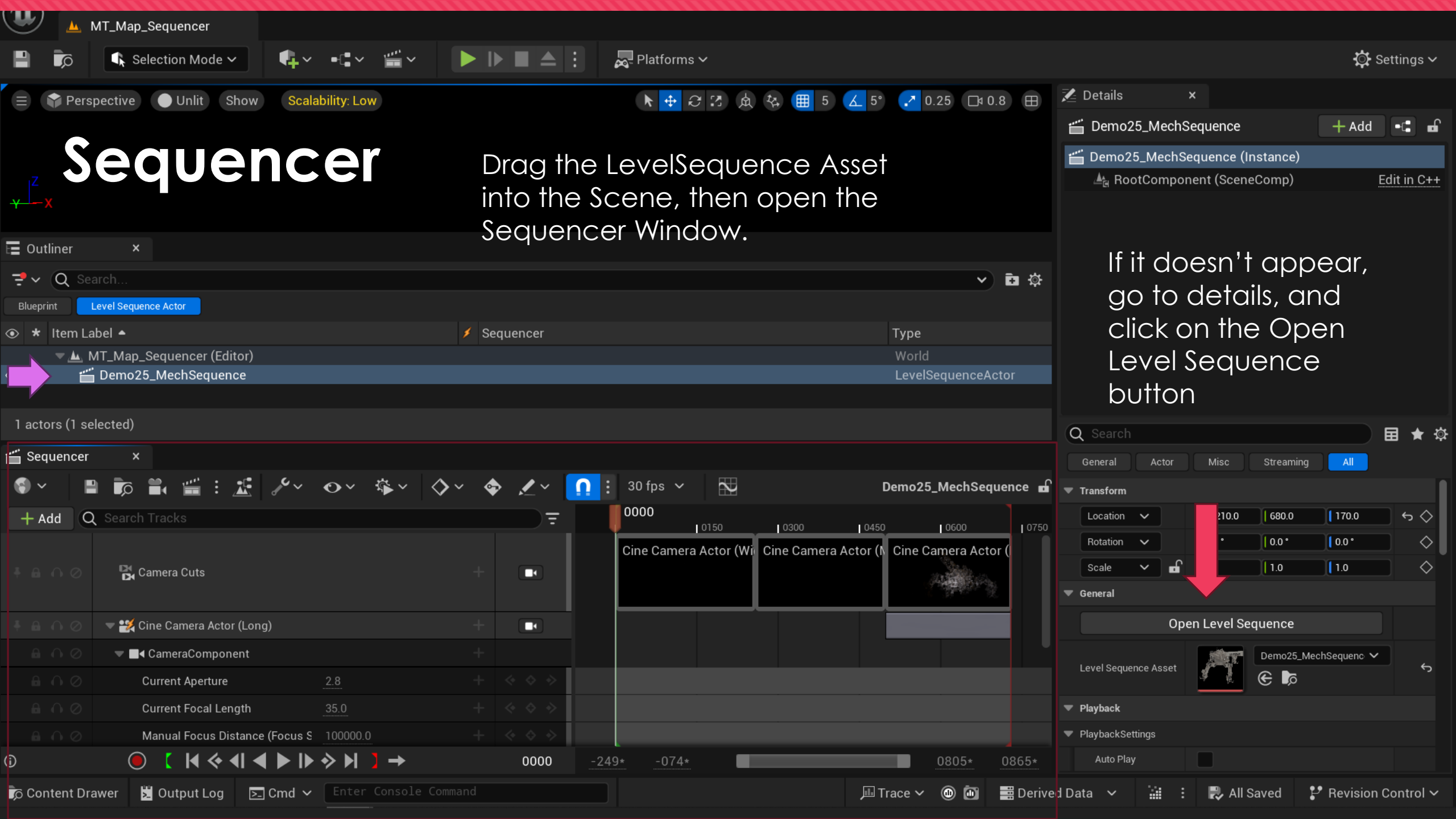
What is Sequencer?

- Sequencer is a module within Unreal for handling animations, cinematic movement, and cameras.
- It can be used to create cutscenes, cinematic sequences to then render out with the Render Queue or allow you to manage multiple cameras.

Sequencer

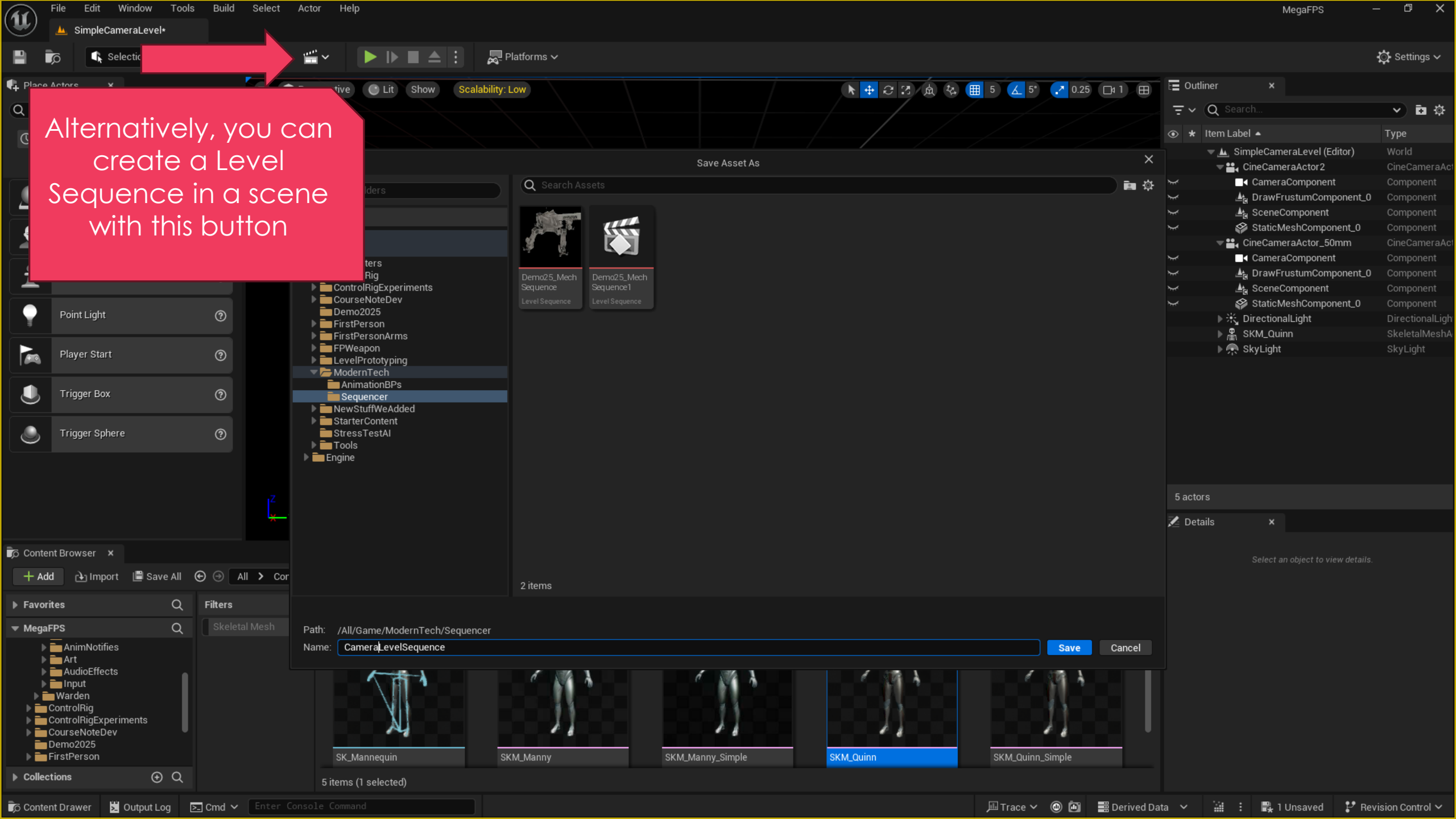
To get started, create a Level Sequence Asset.





Drag the LevelSequence Asset into the Scene, then open the Sequencer Window.

If it doesn't appear, go to details, and click on the Open Level Sequence button



Alternatively, you can
create a Level
Sequence in a scene
with this button



Path: /All/Game/ModernTech/Sequencer
Name: Camera_LevelSequence

Save Cancel



5 items (1 selected)

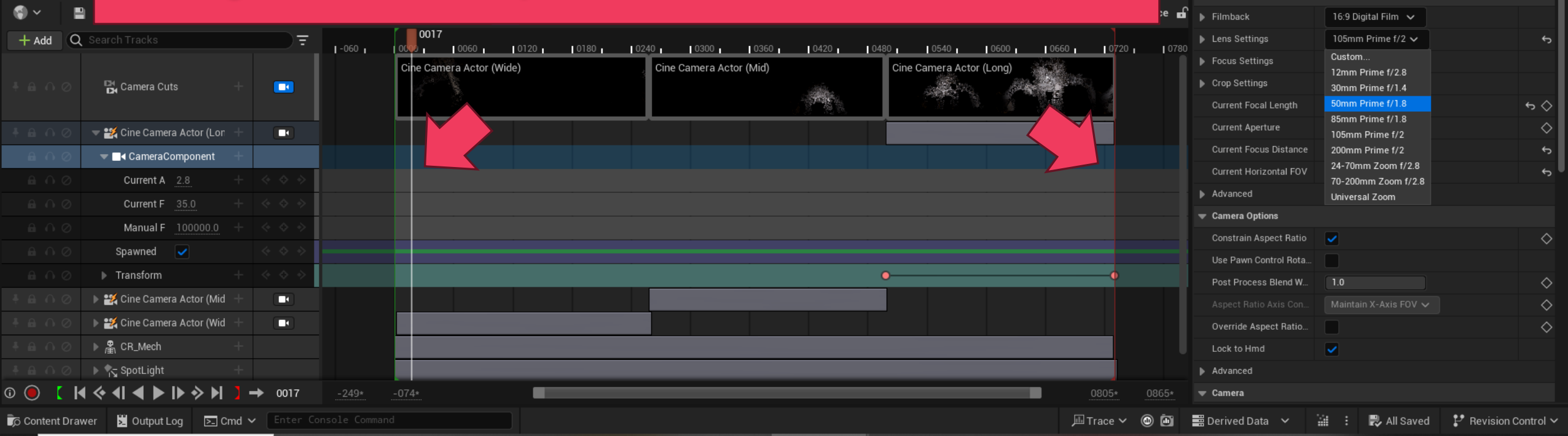
Sequencer Window

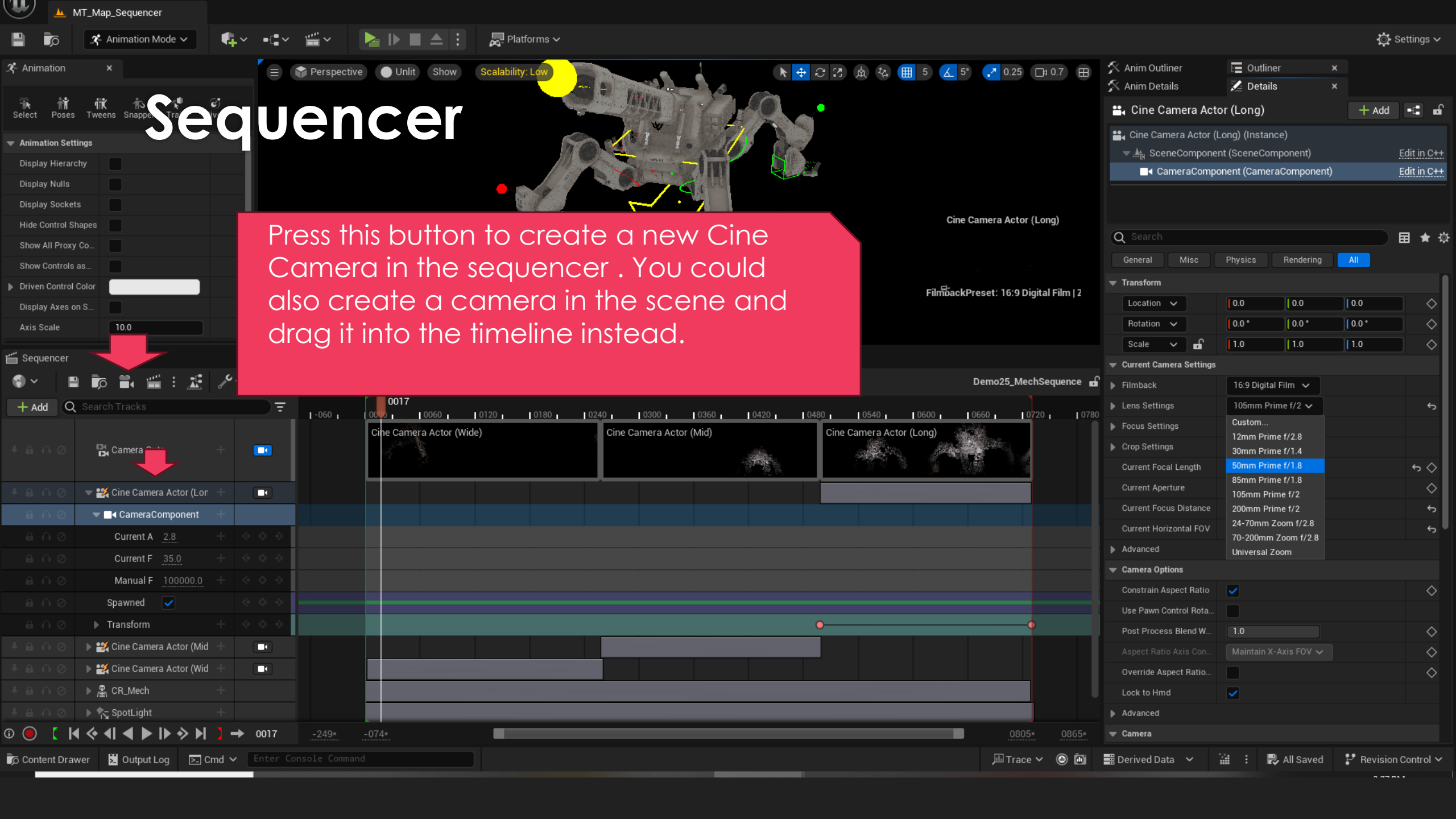
The timeline starts at frame 0, and plays left to right.

The White line is the Playhead and represents the current time that is being played.

You can scrub through an animation by dragging it left and right.

The Red line at the end of the timeline represents the point where playing the sequence will end. You can drag it further to the right to extend the sequence.



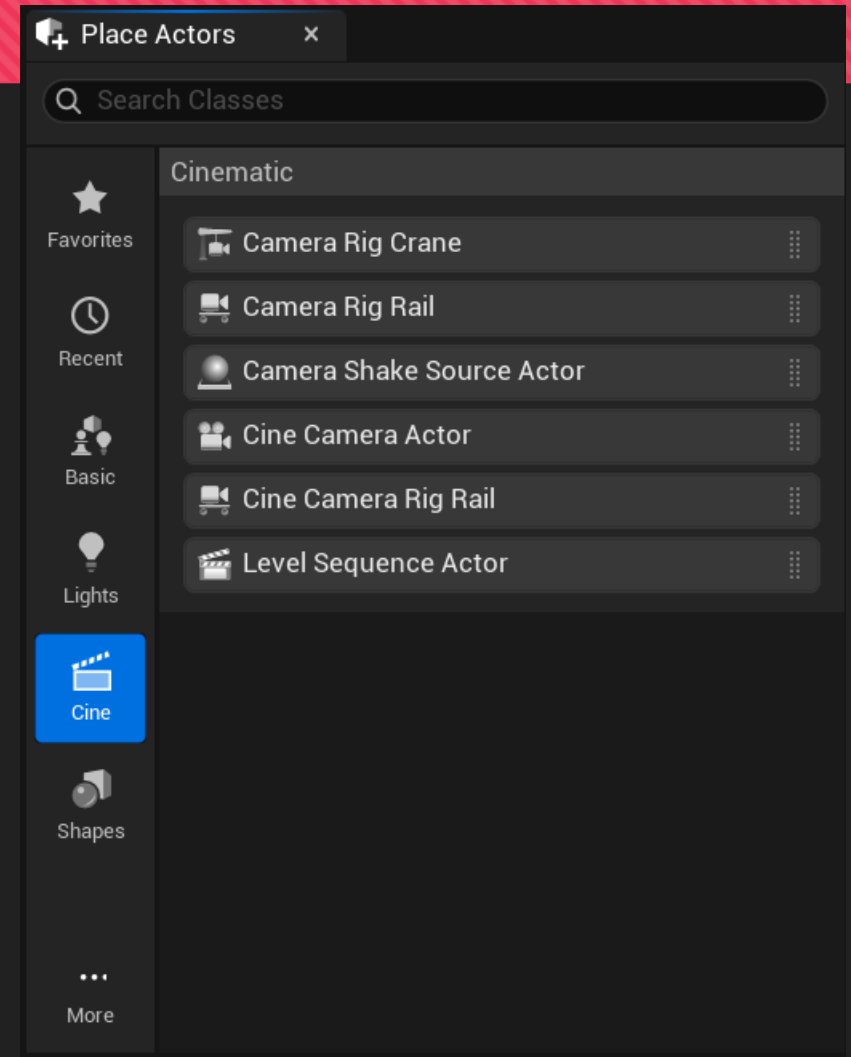


Sequencer

Press this button to create a new Cine Camera in the sequencer . You could also create a camera in the scene and drag it into the timeline instead.

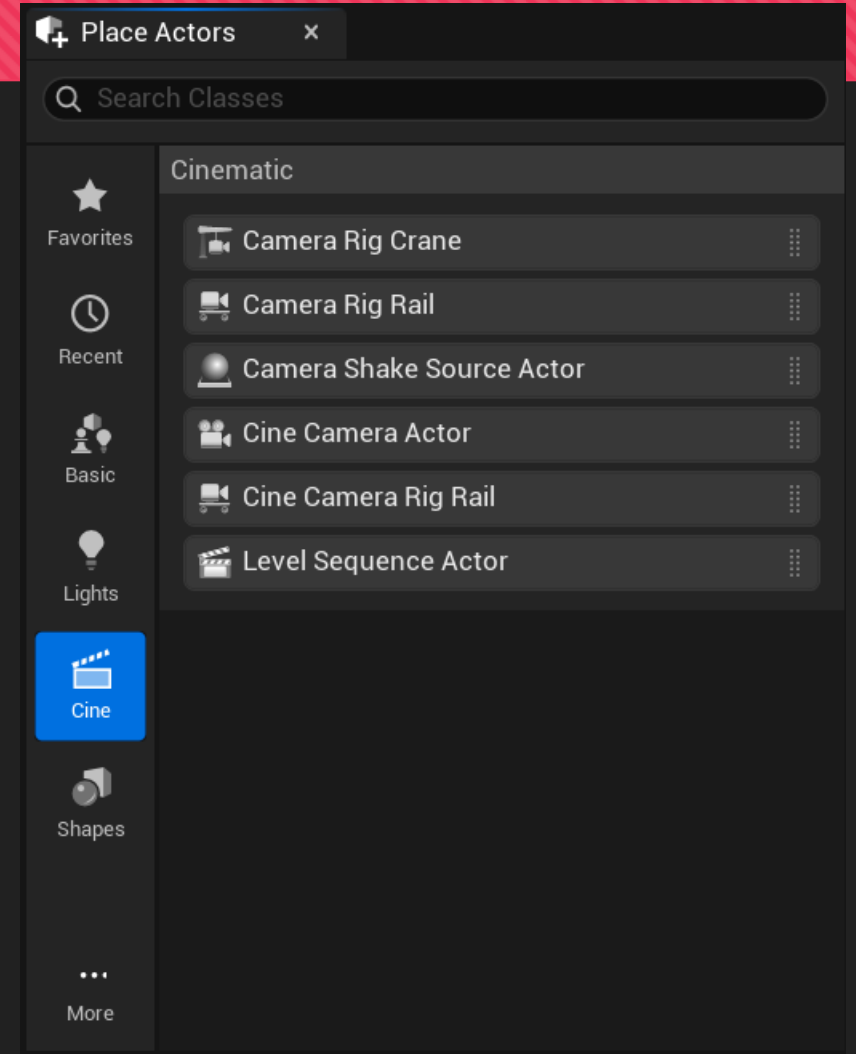
Camera Basics

- There are two types of Camera object in UE, the “regular” Camera, and the Cine Camera.
- They work in a similar way, rendering out to the game view what they can “see”, however the Cine Camera has its settings in a more “traditional” camera context,



Cine Camera Basics

- The Cine Camera is a more advanced Camera that more closely mimics a real camera.
- It can be found in the Place Actors > Cinematic section.



CameraActor

CameraActor (Instance)

SceneComponent (SceneComponent)

CameraComponent (CameraComponent)

Search

GeneralActorCameraCamera OptionsMiscPost ProcessStreaming

All

Mobility

StaticStationaryMovable

Camera Settings

Projection Mode

Perspective

Field Of View

90.0 °

Aspect Ratio

1.77778

width x

Overscan

First Person Field Of View

90.0 °

First Person Scale

1.0

Camera Options

Constrain Aspect Ratio

☒

Use Pawn Control Rotation

☐

Post Process Blend Weight

1.0

Aspect Ratio Axis Constraint

Maintain X-Axis FOV

Override Aspect Ratio Axis Constraint

☐

Lock to Hmd

☒

“Regular”

CineCameraActor

CineCameraActor (Instance)

SceneComponent (SceneComponent)

CameraComponent (CameraComponent)

Search

GeneralCameraCamera OptionsCurrent Camera SettingsMiscPhysics

Post ProcessRenderingAll

Current Camera Settings

Overscan

Filmback

16:9 Digital Film

Lens Settings

Universal Zoom

Focus Settings

Focus Method

Manual

Manual Focus Distance

100000.0 cm

Draw Debug Focus Plane

☐

Debug Focus Plane Color

Smooth Focus Changes

☐

Focus Smoothing Interp Speed

8.0

Focus Offset

0.0

Crop Settings

No Crop

Current Focal Length

35.0

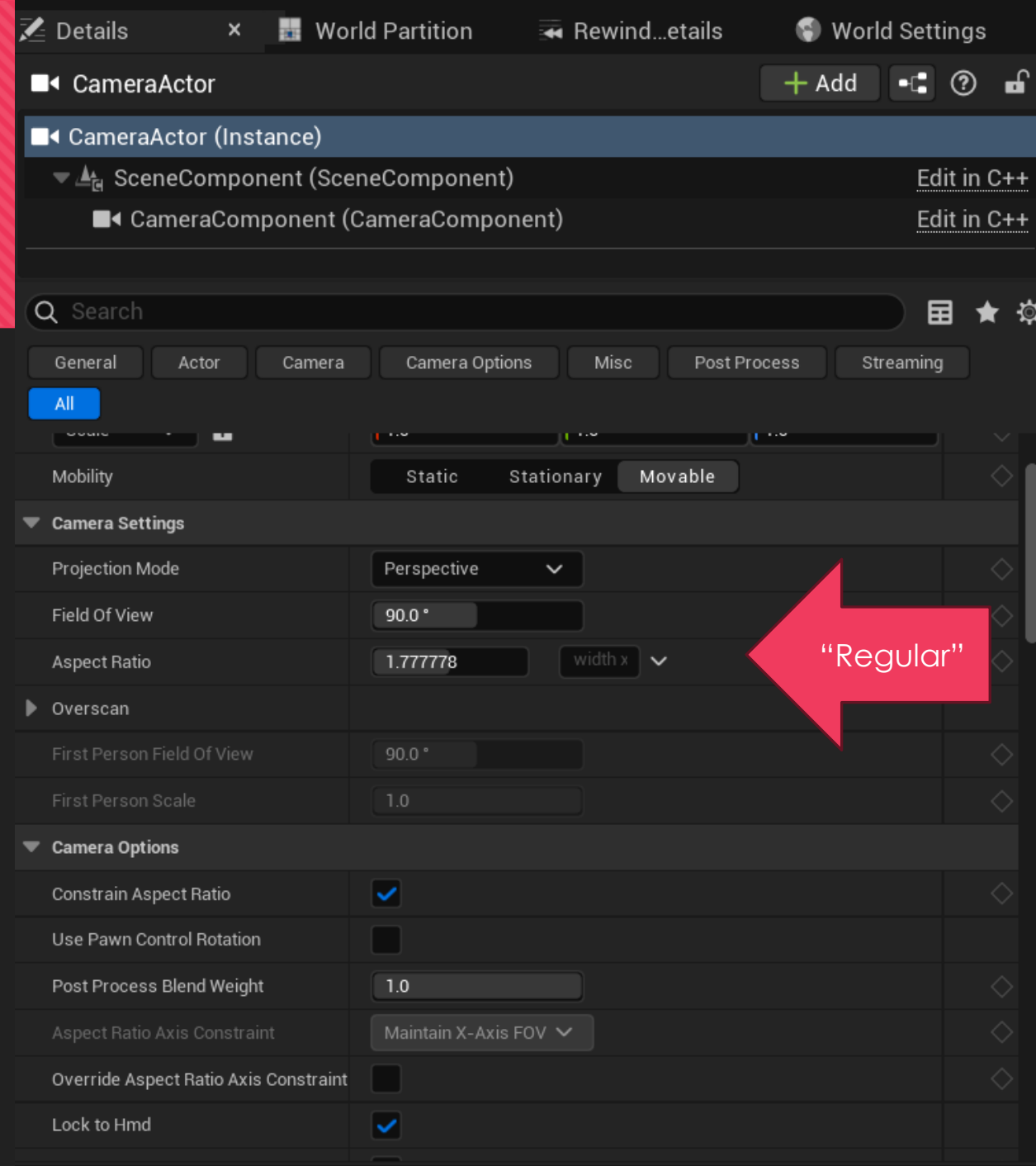
Current Aperture

2.8

“Cine Camera”

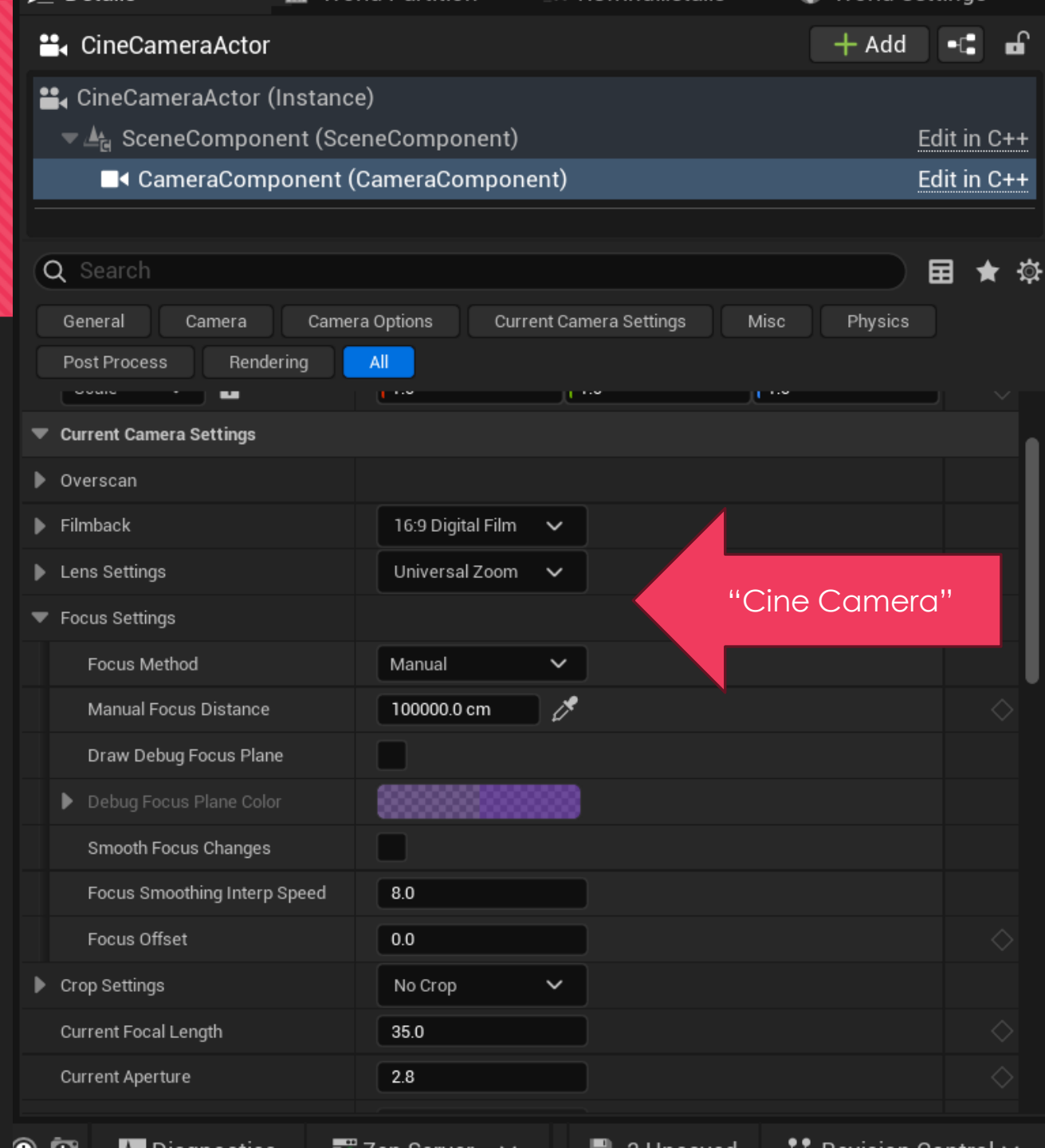
Camera Basics

- The regular camera is what you are used to in most game engines.
- It has the Field of View measured as an angle, which is typical for game engines.
- There are otherwise somewhat limited settings to control it, at least compared to the Cine Camera



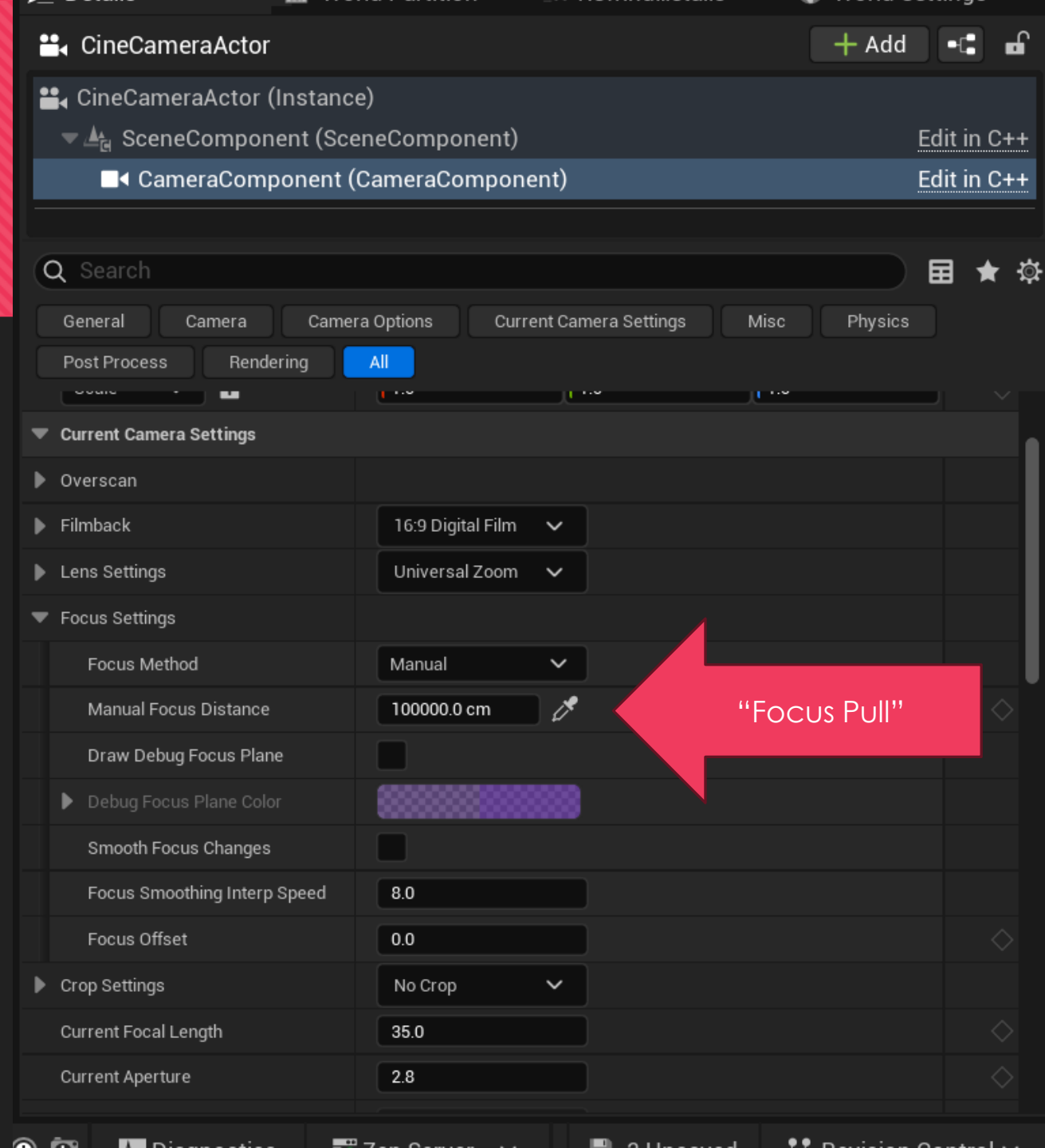
Cine Camera Basics

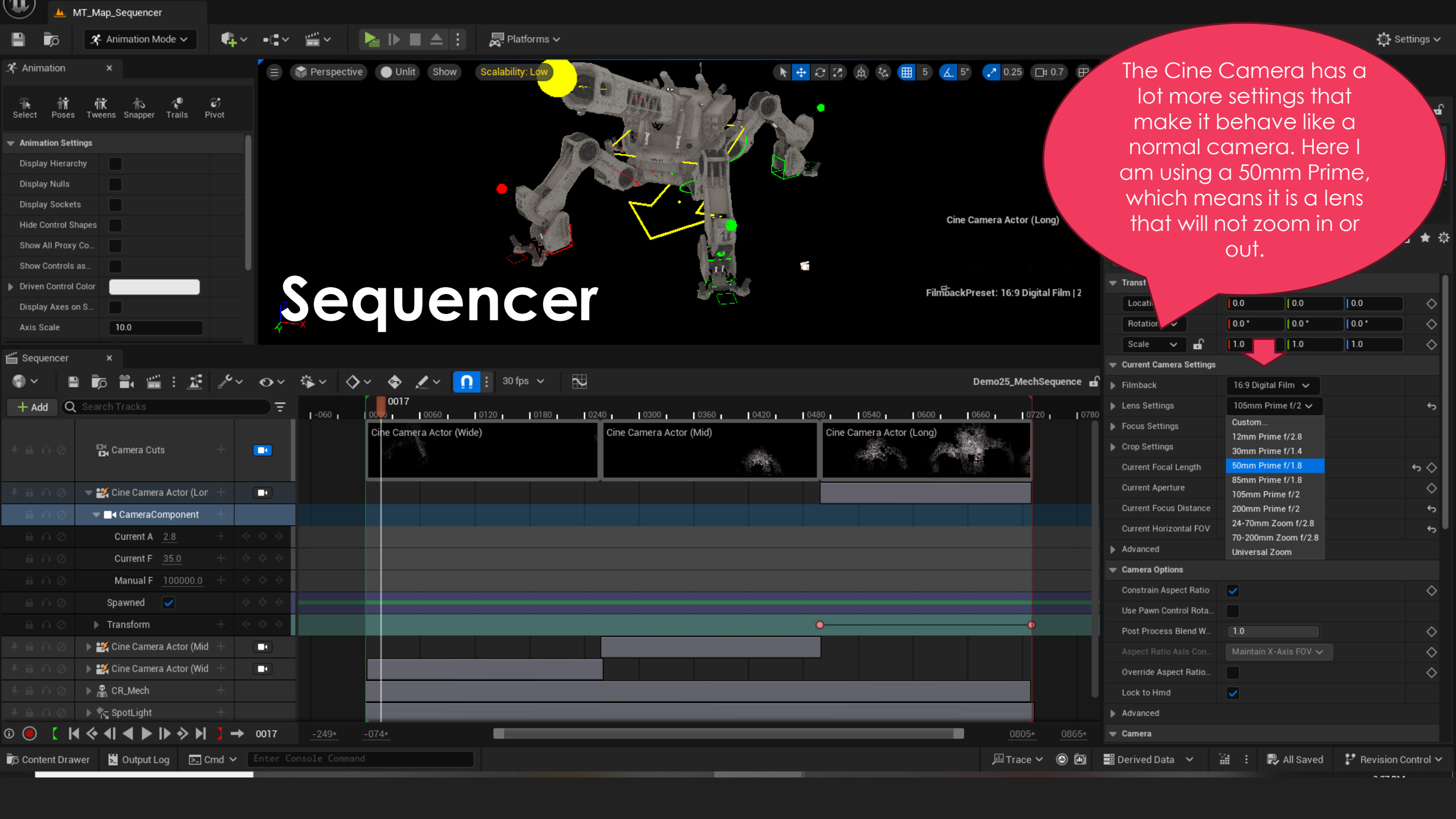
- The Cine Camera is more what a Cinematographer working with a “real” physical camera would expect to see. It might be a bit confusing at first, but there are some cool features to explore.
- Filmback: This is talking about the specific type of sensor and aspect ratio. This would be equivalent to swapping the “Body” of a physical camera.
- Lens Settings: This gives a lot more control and options for handling the zoom level of the lens and mimic how a real camera would work.



Cine Camera Basics

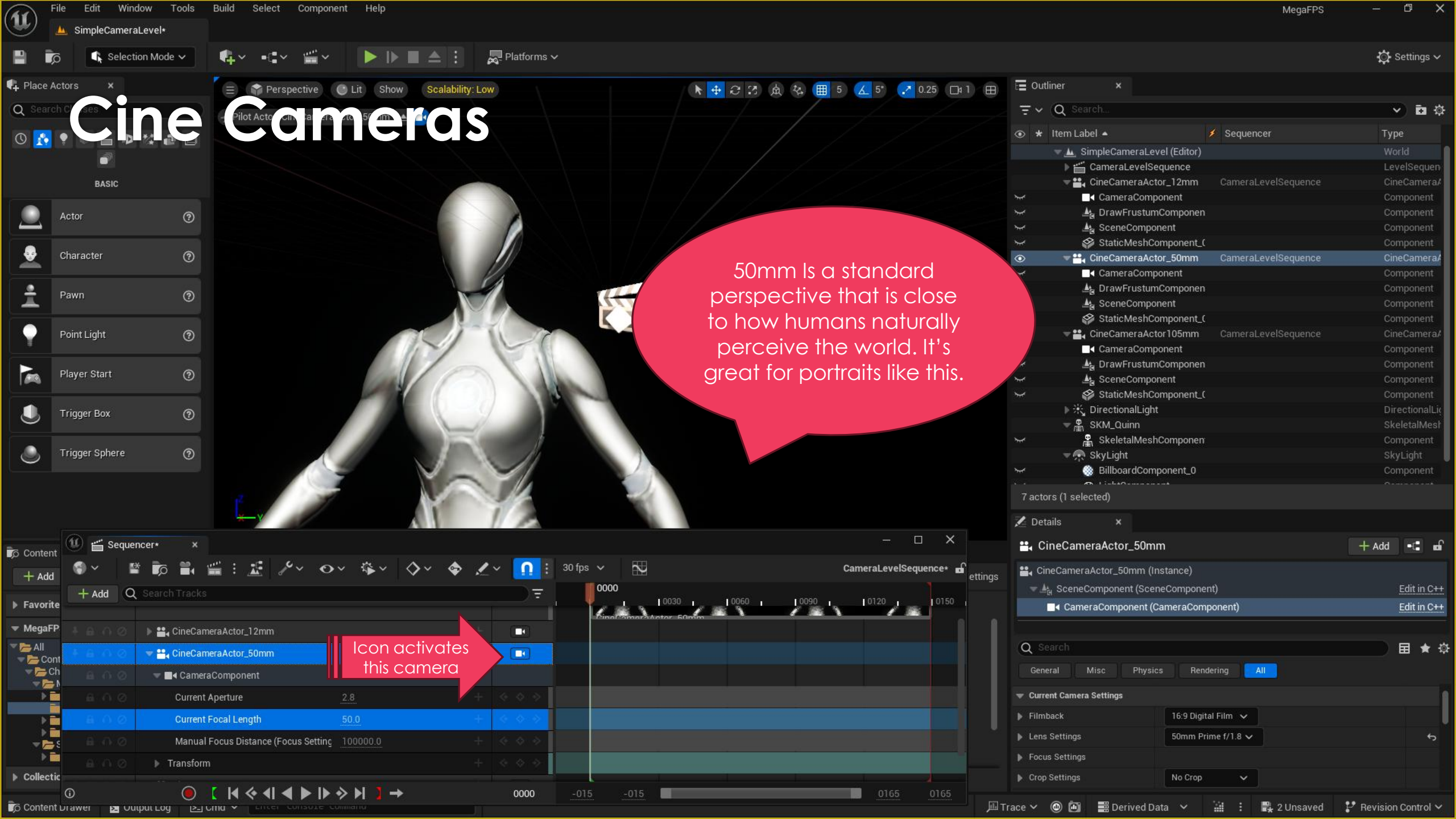
- A cool thing you can do with the Cine Camera is actually sample an object in the scene to get the correct focal length.
- With a physical camera, most things in the shot will not be in focus, background elements for example will feel soft and blurred, and this can help focus a viewers attention.
- The eye dropper tool will allow you to “pick” a focal point for your lens to focus on.
- Note: if you move the camera, your target will get out of focus, but we’ll talk more about how to deal with that later.





Sequencer

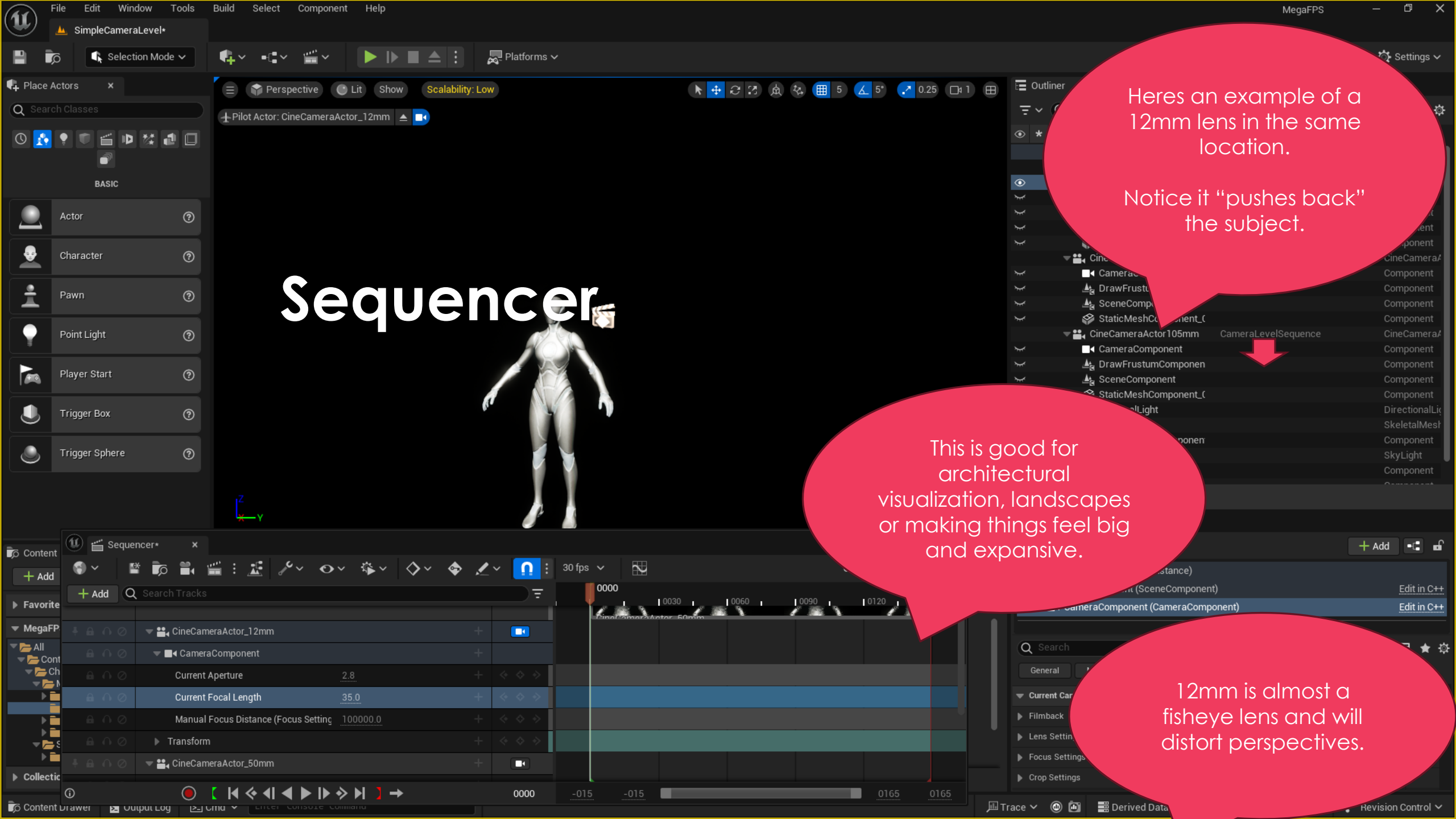
The Cine Camera has a lot more settings that make it behave like a normal camera. Here I am using a 50mm Prime, which means it is a lens that will not zoom in or out.



Cine Cameras

50mm Is a standard perspective that is close to how humans naturally perceive the world. It's great for portraits like this.

Icon activates this camera



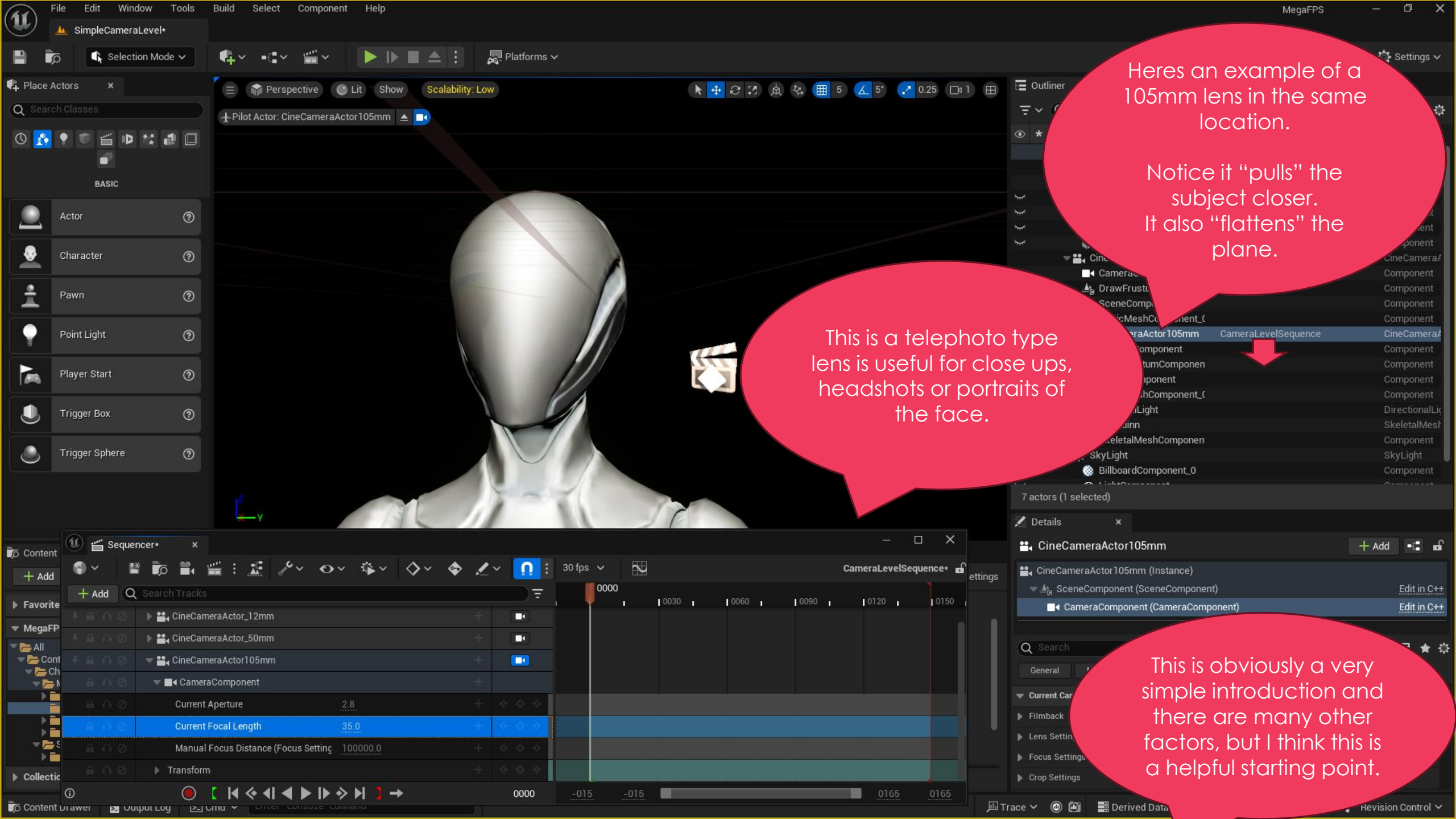
Sequencer

Here's an example of a 12mm lens in the same location.

Notice it "pushes back" the subject.

This is good for architectural visualization, landscapes or making things feel big and expansive.

12mm is almost a fisheye lens and will distort perspectives.



Heres an example of a 105mm lens in the same location.

Notice it "pulls" the subject closer. It also "flattens" the plane.

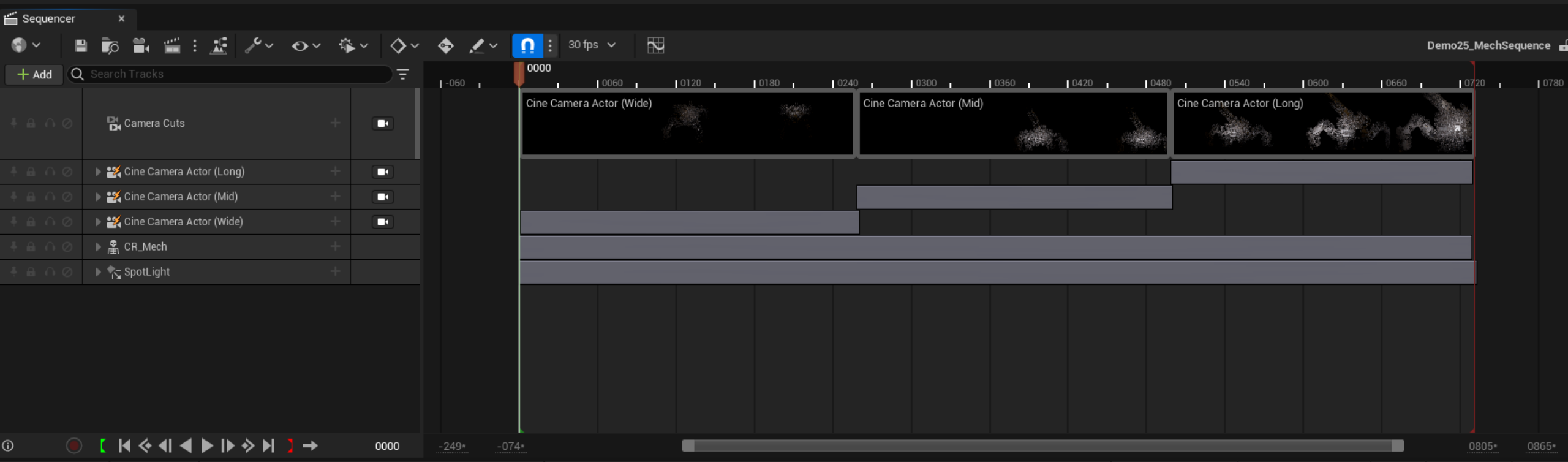
This is a telephoto type lens is useful for close ups, headshots or portraits of the face.

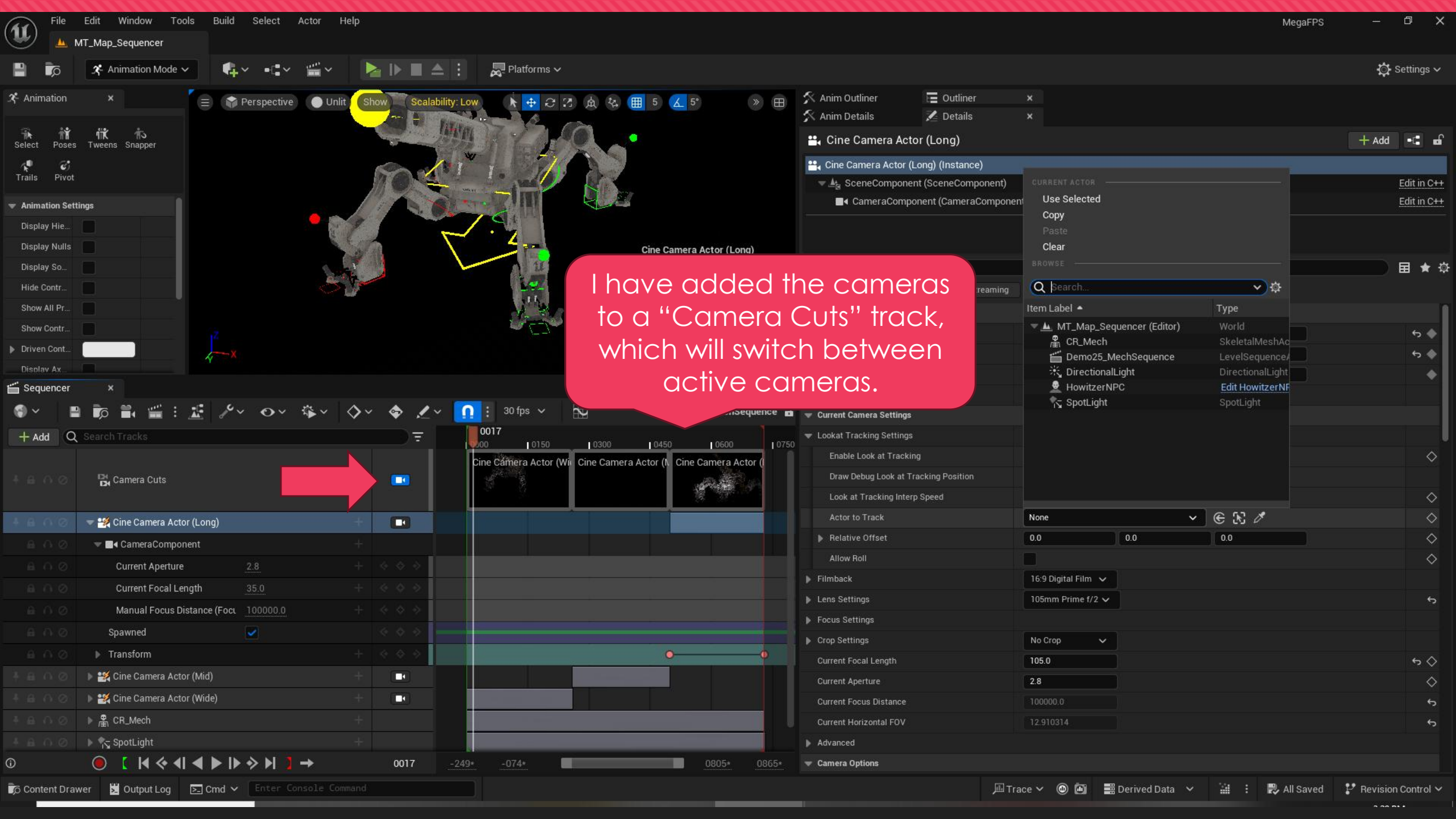
This is obviously a very simple introduction and there are many other factors, but I think this is a helpful starting point.

Selecting between cameras

Sequencer Timeline

The Sequencer timeline itself is modelled after non-linear editing software like you would use to make a movie, so it requires a little bit of film / editing knowledge and terminology. Basically though, you can assign your camera actor to the cut track, then edit the cuts (between cameras) like you would edit a movie.





I have added the cameras to a "Camera Cuts" track, which will switch between active cameras.

Sequencer: More Resources

Required Reading for this Module:

<https://dev.epicgames.com/documentation/en-us/unreal-engine/cinematics-and-movie-making-in-unreal-engine>

For more information on how to edit tracks using the Sequencer, these are useful. ->

Sequencer Editor

These pages describe tools, features, and workflows of the Sequencer Editor and tools related to Sequencer.



Sequencer Overview

Learn about Level Sequences and the primary features of the Sequencer Editor.



Sequencer Editor

An overview of the Sequence Editor's user interface, tools, and options.



Curve Editor

Tweak your keyframes and curves by using the Curve Editor and tools within it.



Tracks

Create tracks that affect your actors in Sequencer.



Sequences, Shots, and Takes

Edit cinematics in a non-linear editor using Sequences, Shots, and Takes.



Actor Sequence Component

Describes how you can embed Sequences within Actor Blueprints using an Actor Sequence Component.

Starting Out

Sequencer Editor

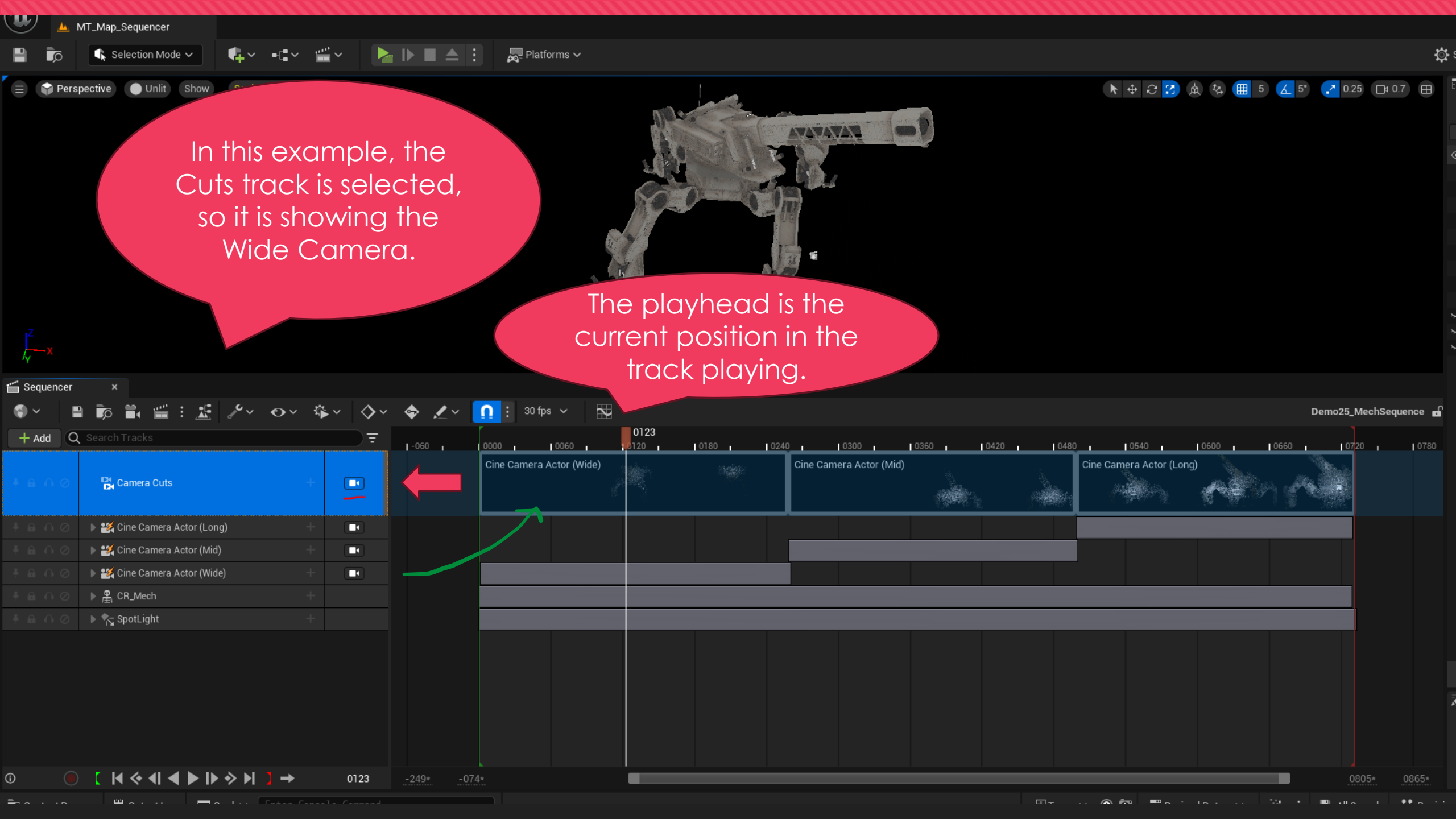
Cameras

Rendering

Workflow Guides and Examples

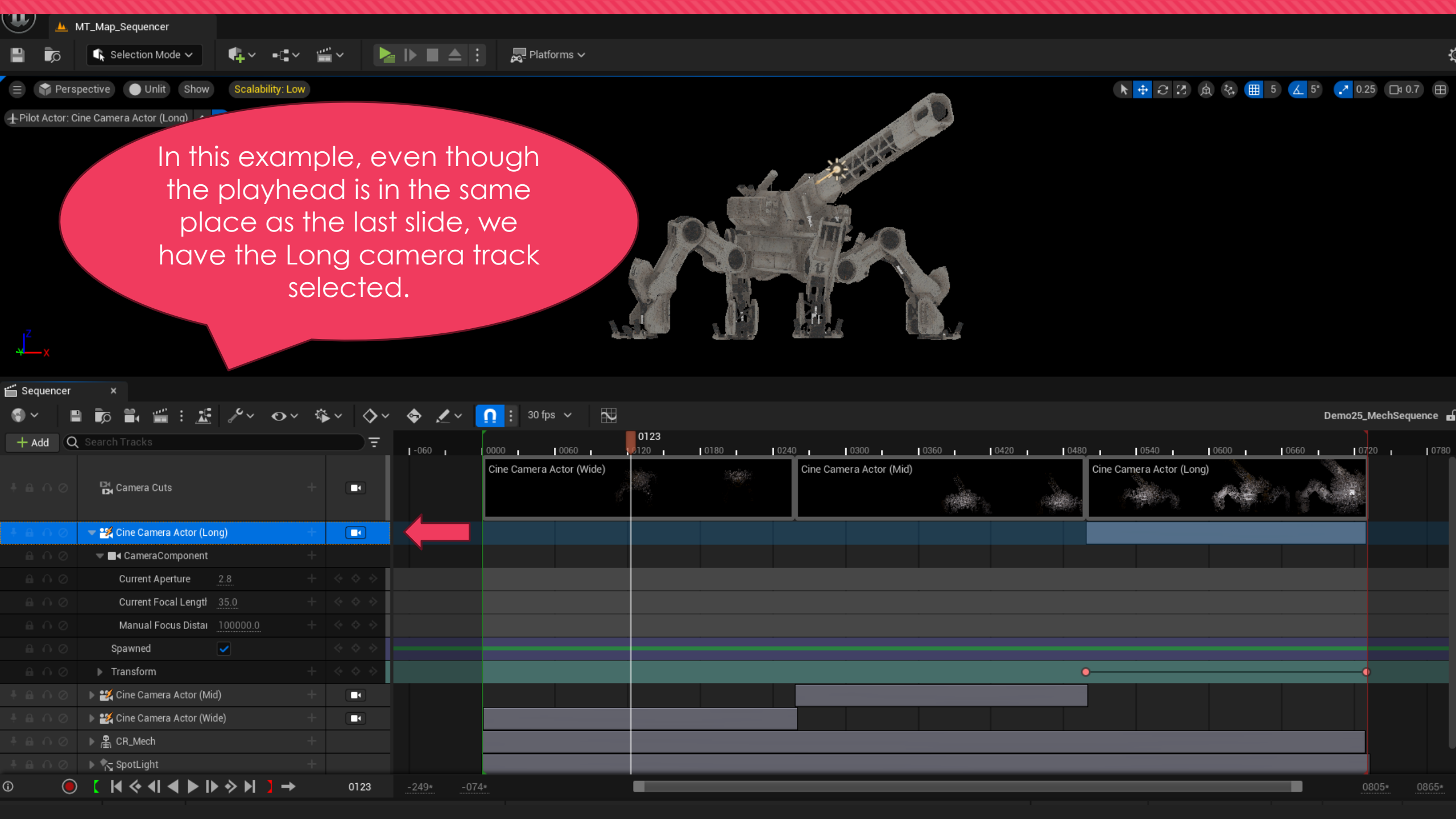
Cheat Sheet and Workflow shortcuts

Cinematic Samples



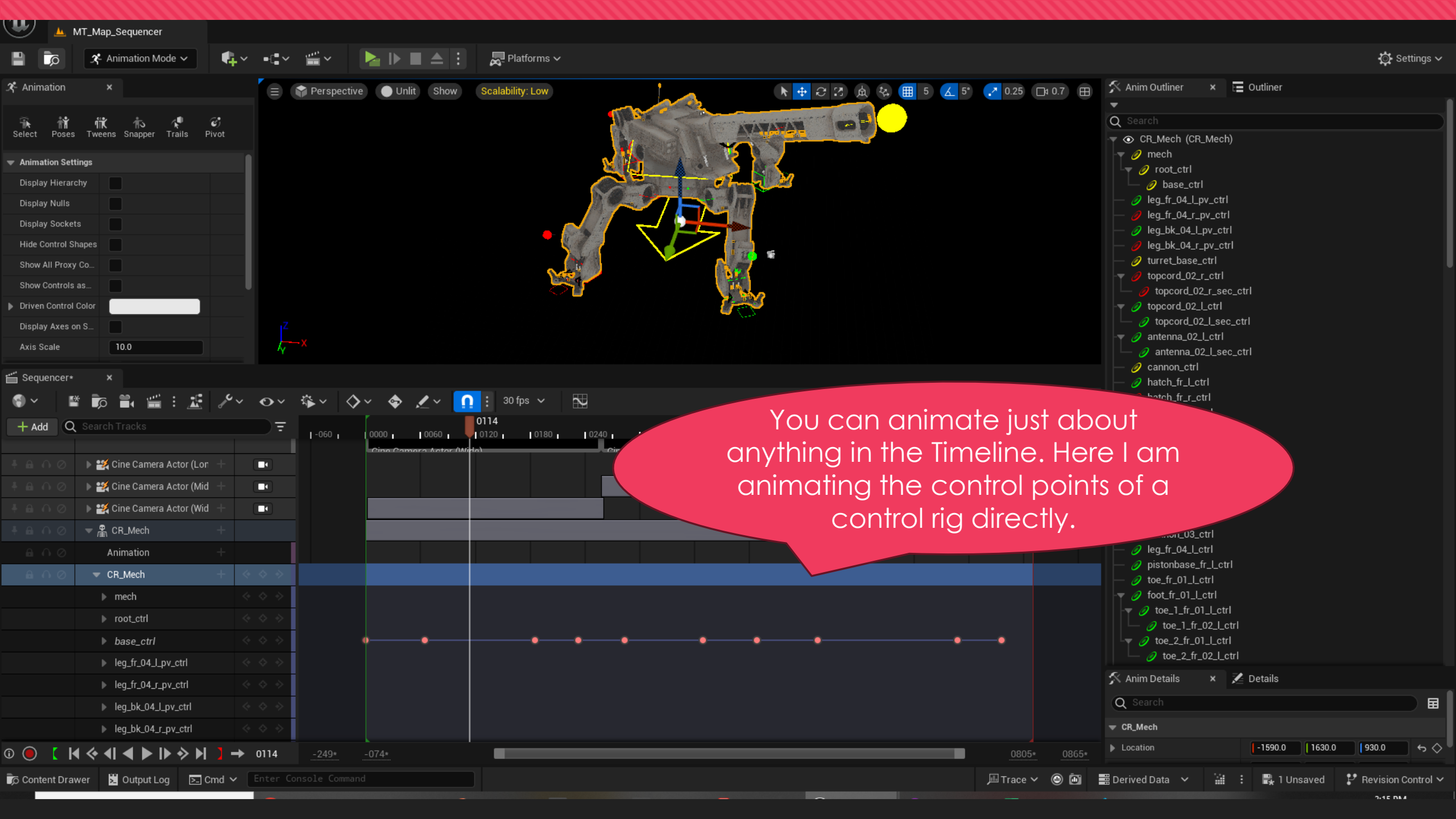
In this example, the Cuts track is selected, so it is showing the Wide Camera.

The playhead is the current position in the track playing.

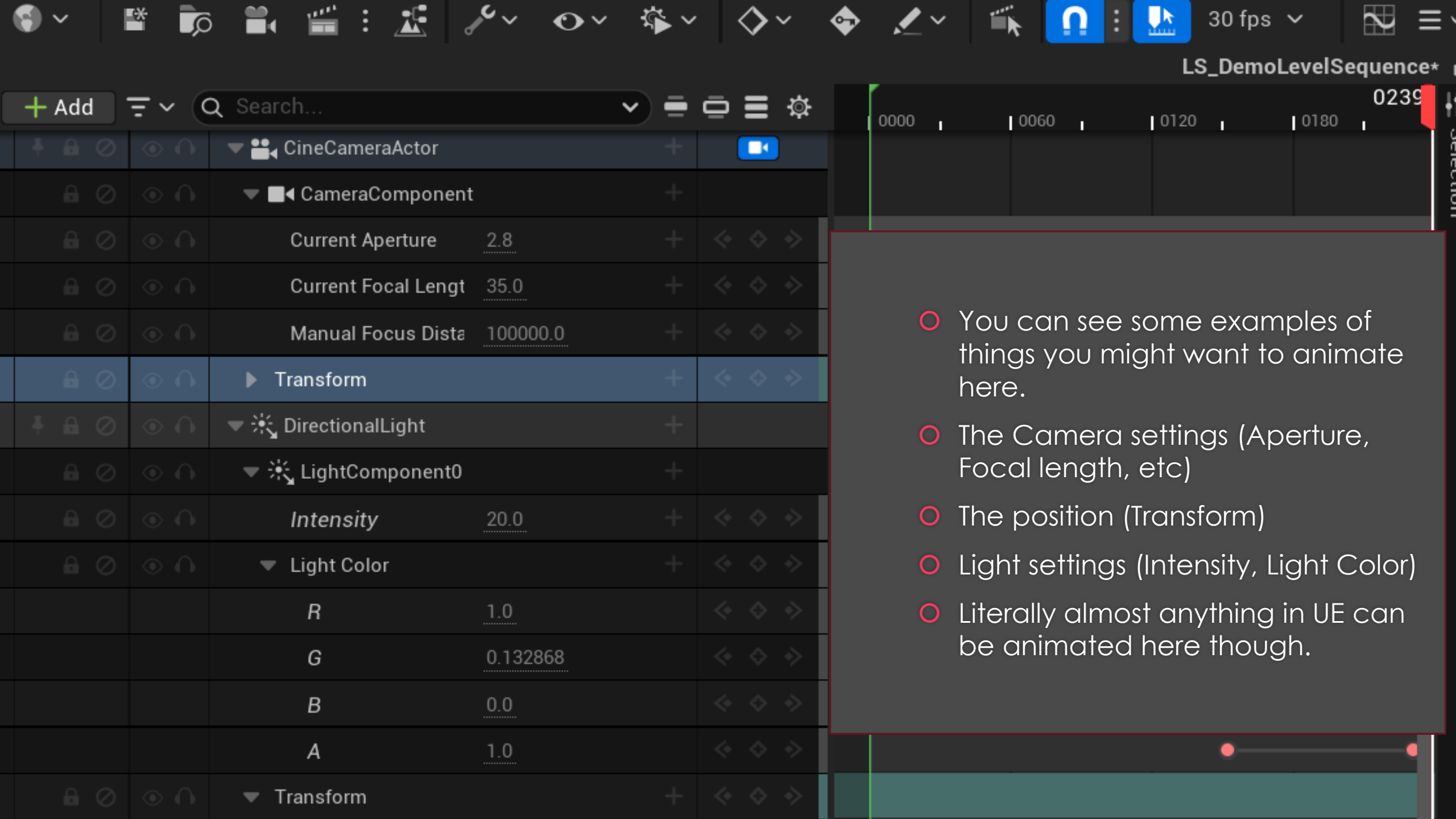


In this example, even though the playhead is in the same place as the last slide, we have the Long camera track selected.

Animating in the Timeline

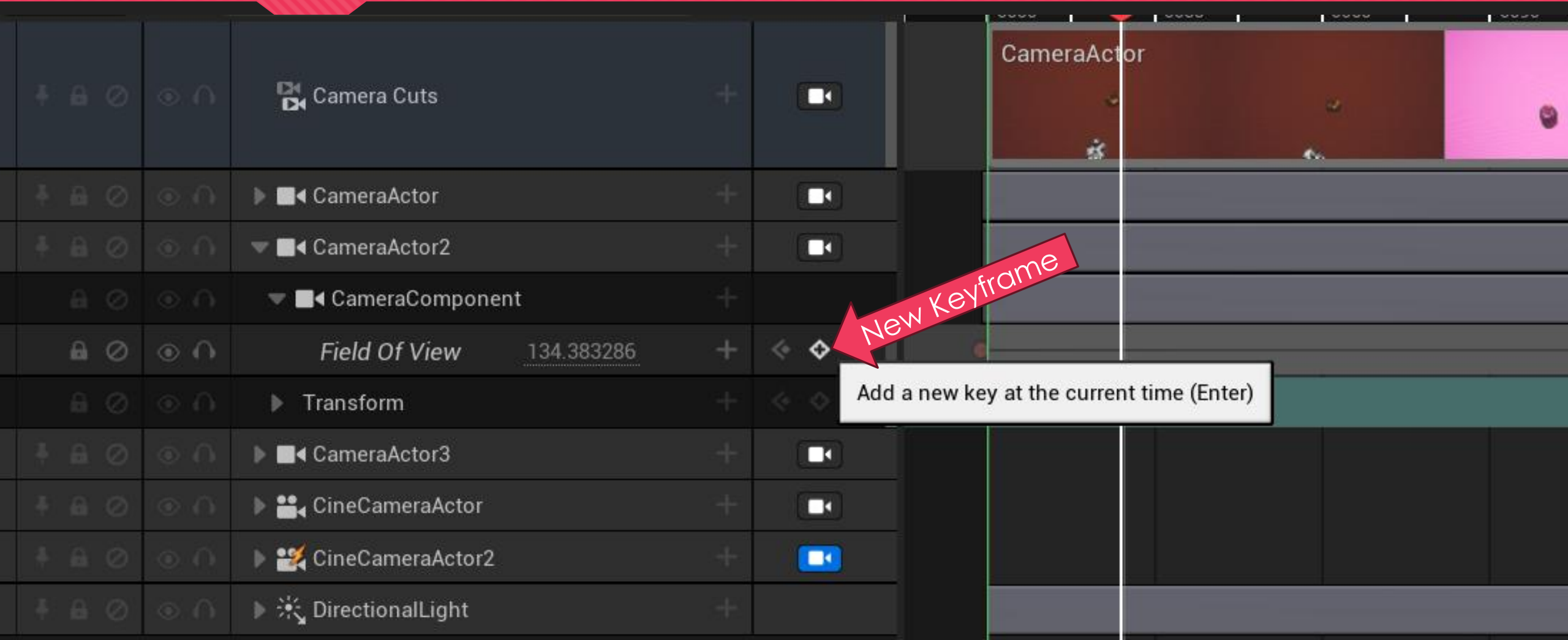


You can animate just about anything in the Timeline. Here I am animating the control points of a control rig directly.



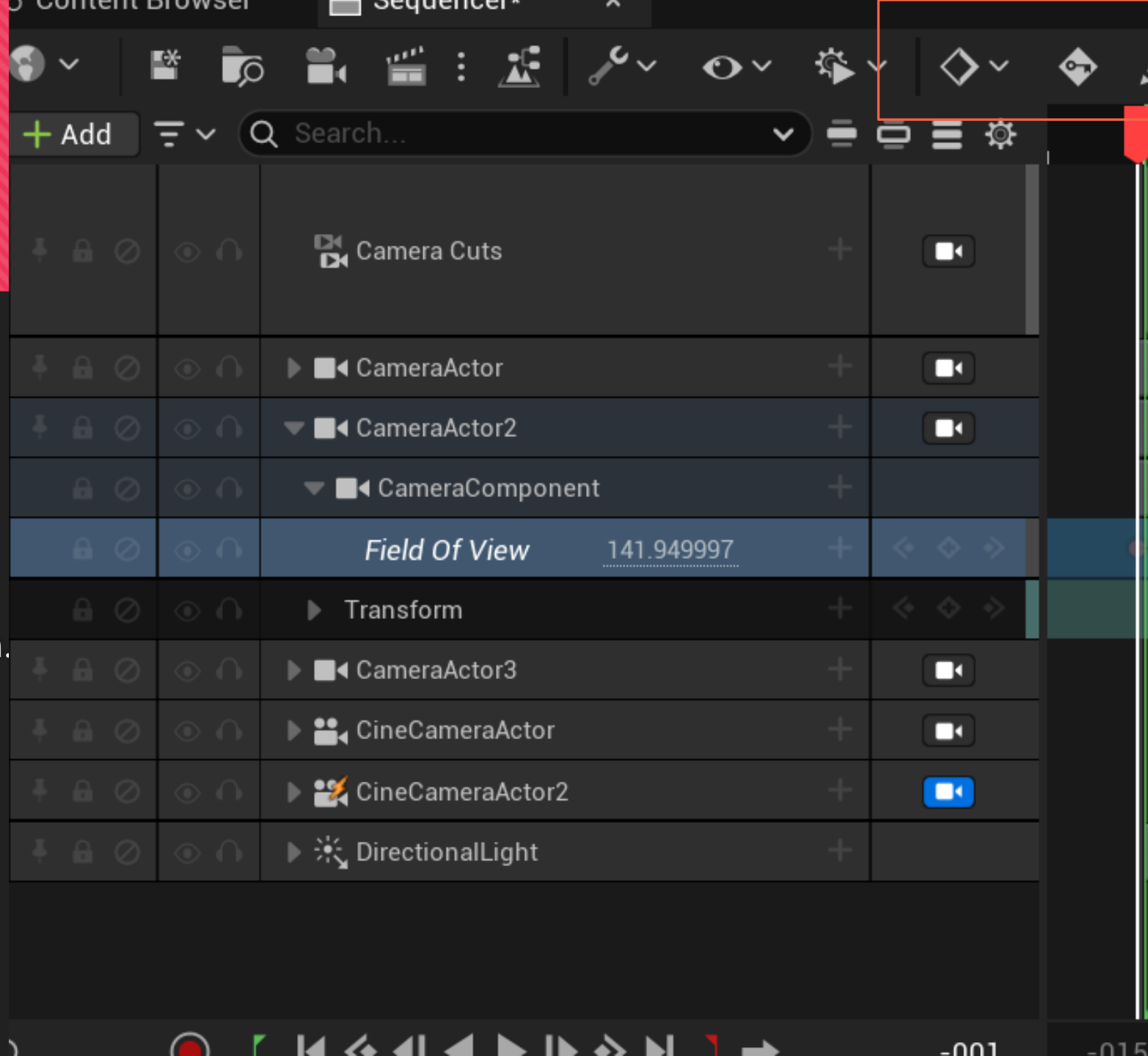
- You can see some examples of things you might want to animate here.
- The Camera settings (Aperture, Focal length, etc)
- The position (Transform)
- Light settings (Intensity, Light Color)
- Literally almost anything in UE can be animated here though.

Keyframe basics



Keyframe Automatically

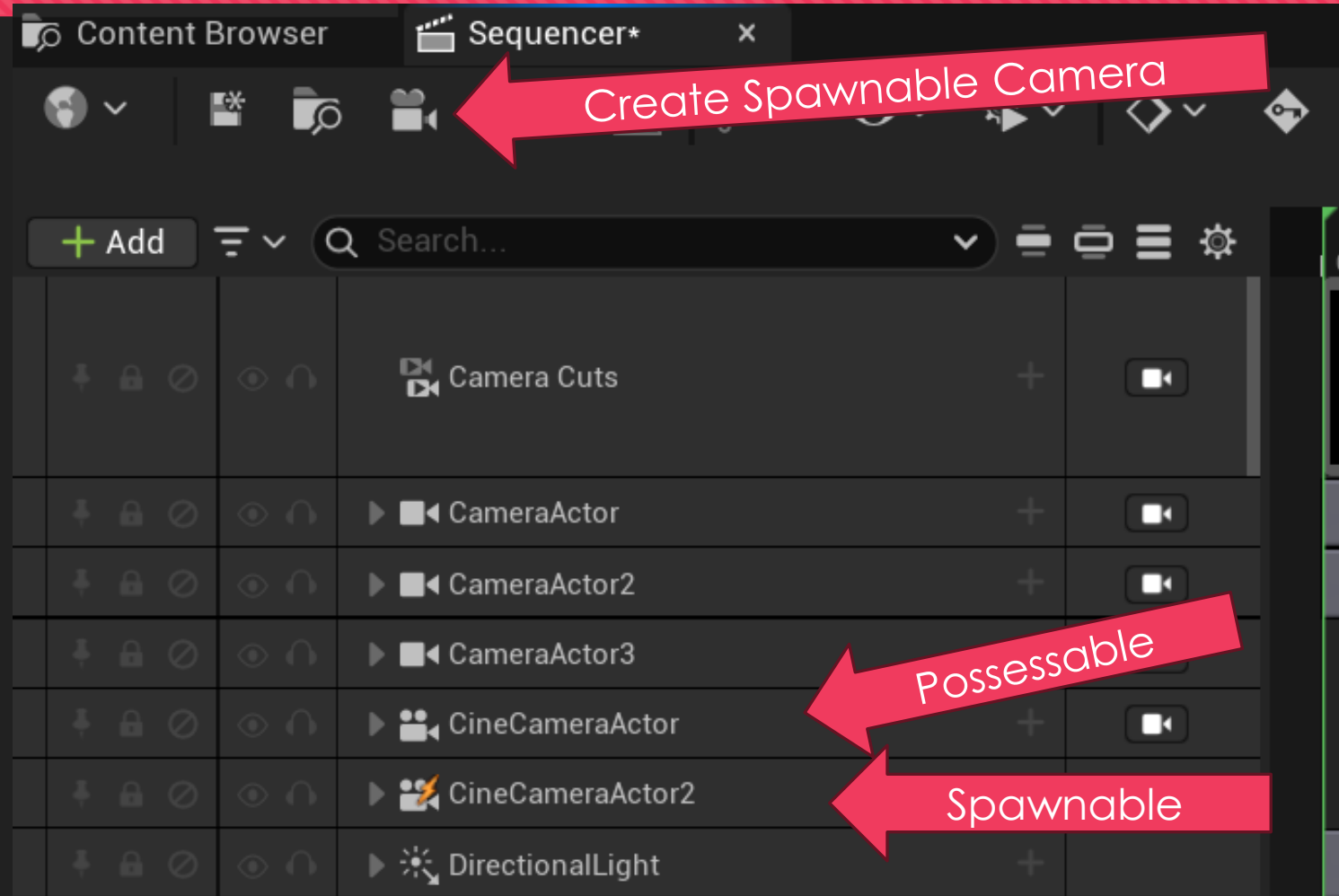
- These settings can “record” changes as keyframes.
- They are similar to how Unity’s animation system will record keyframes. Make sure you are being careful with them though. It’s easy to record things you didn’t anticipate.
- Read the tooltips for context and more detail.



Possessable vs Spawning Actors

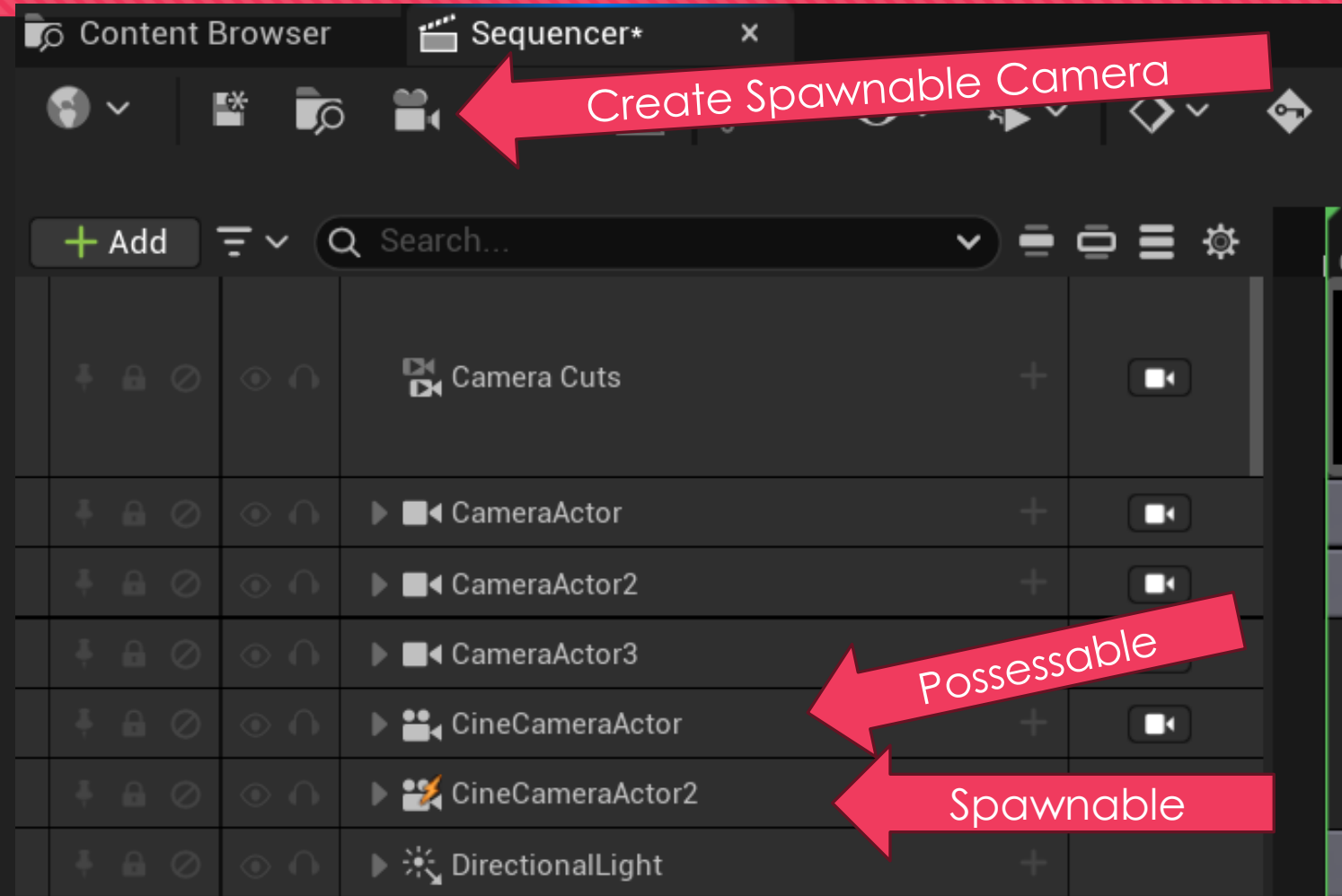
Possessable Objects

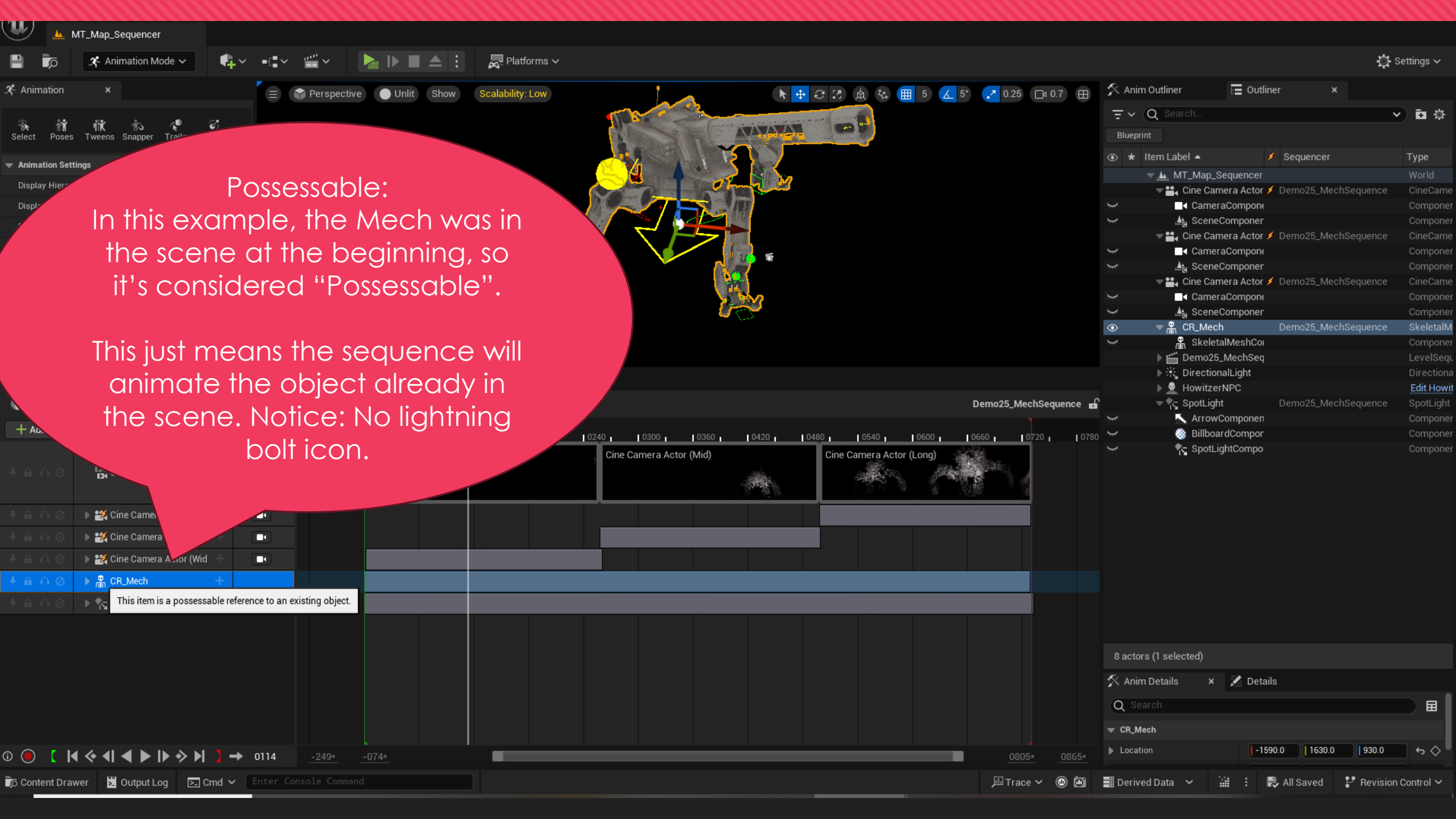
- Depending on how you created your Camera, it might be considered “Possessable” or not.
- If you dropped a camera into your scene, it would exist in the scene at runtime, it would be considered “Possessable”
- If you created your camera using the icon in Sequencer, you will notice it has a little lightning bolt attached.
- This means it is a “Spawnable” object, rather than possessable.



Possessable Objects

- While I won't get deep into it, basically, the Spawnable camera are being created dynamically at runtime when they are needed, so you won't find them in the scene.
- Either will work, it is very confusing to work with cameras if you don't understand the difference though.
- More Info:
<https://dev.epicgames.com/documentation/en-us/unreal-engine/unreal-engine-5-7-documentation>





Possessable:

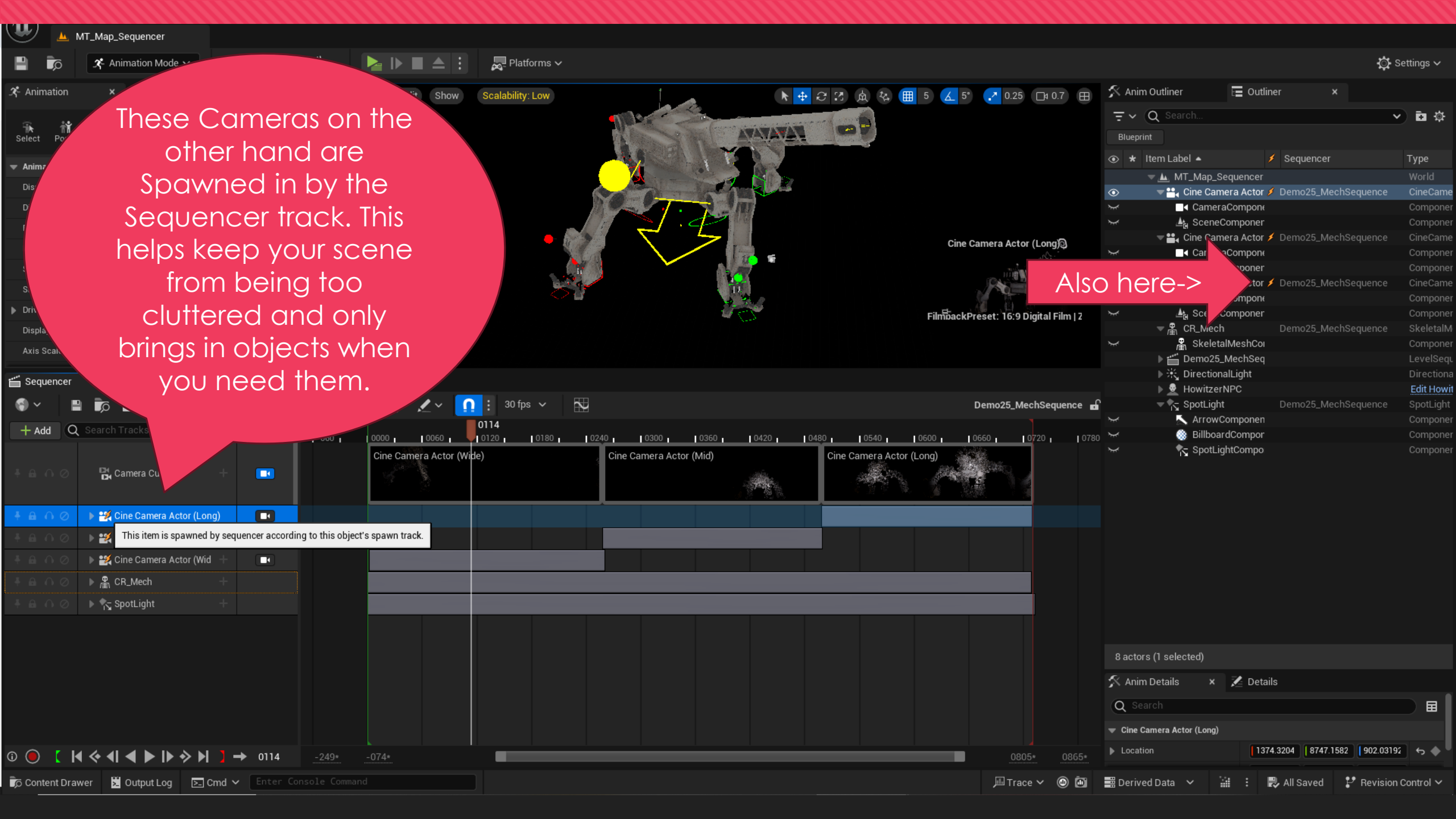
In this example, the Mech was in the scene at the beginning, so it's considered "Possessable".

This just means the sequence will animate the object already in the scene. Notice: No lightning bolt icon.

This item is a possessable reference to an existing object.

These Cameras on the other hand are Spawned in by the Sequencer track. This helps keep your scene from being too cluttered and only brings in objects when you need them.

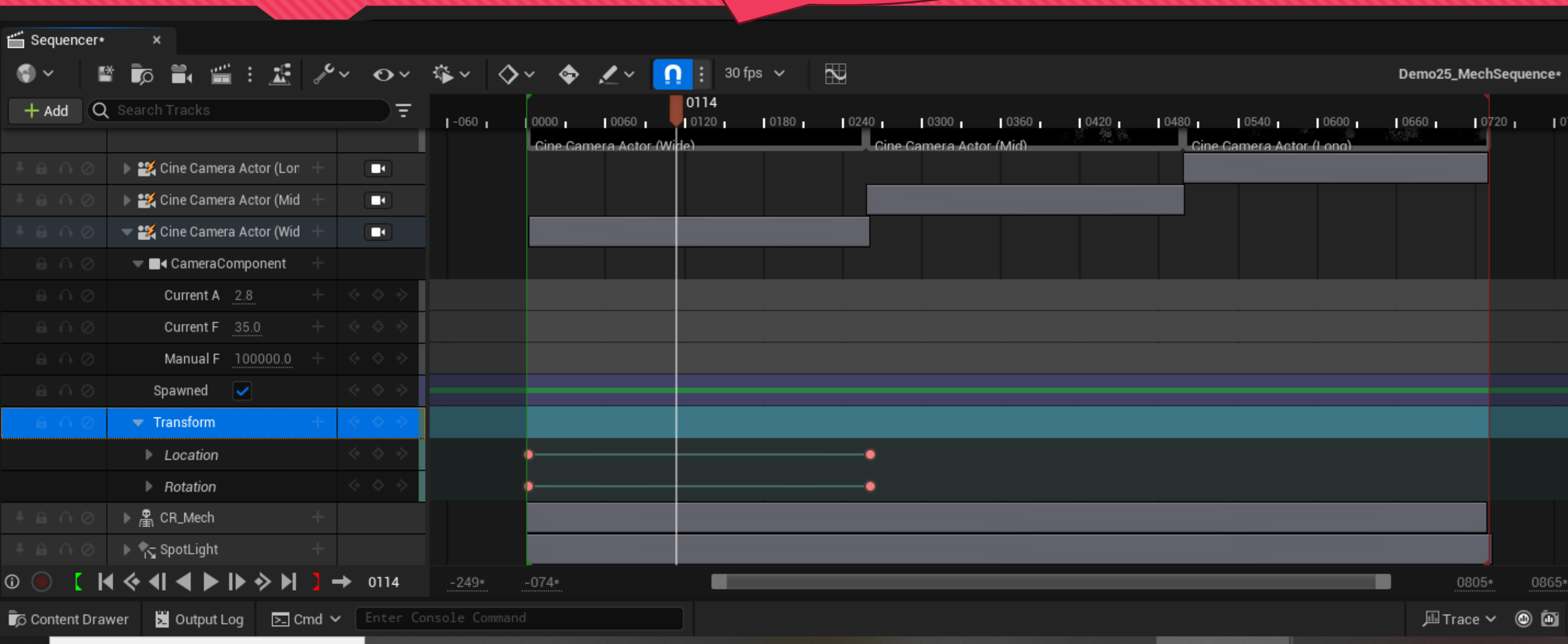
Also here->



Sequencer: Misc Tips

Sequencer

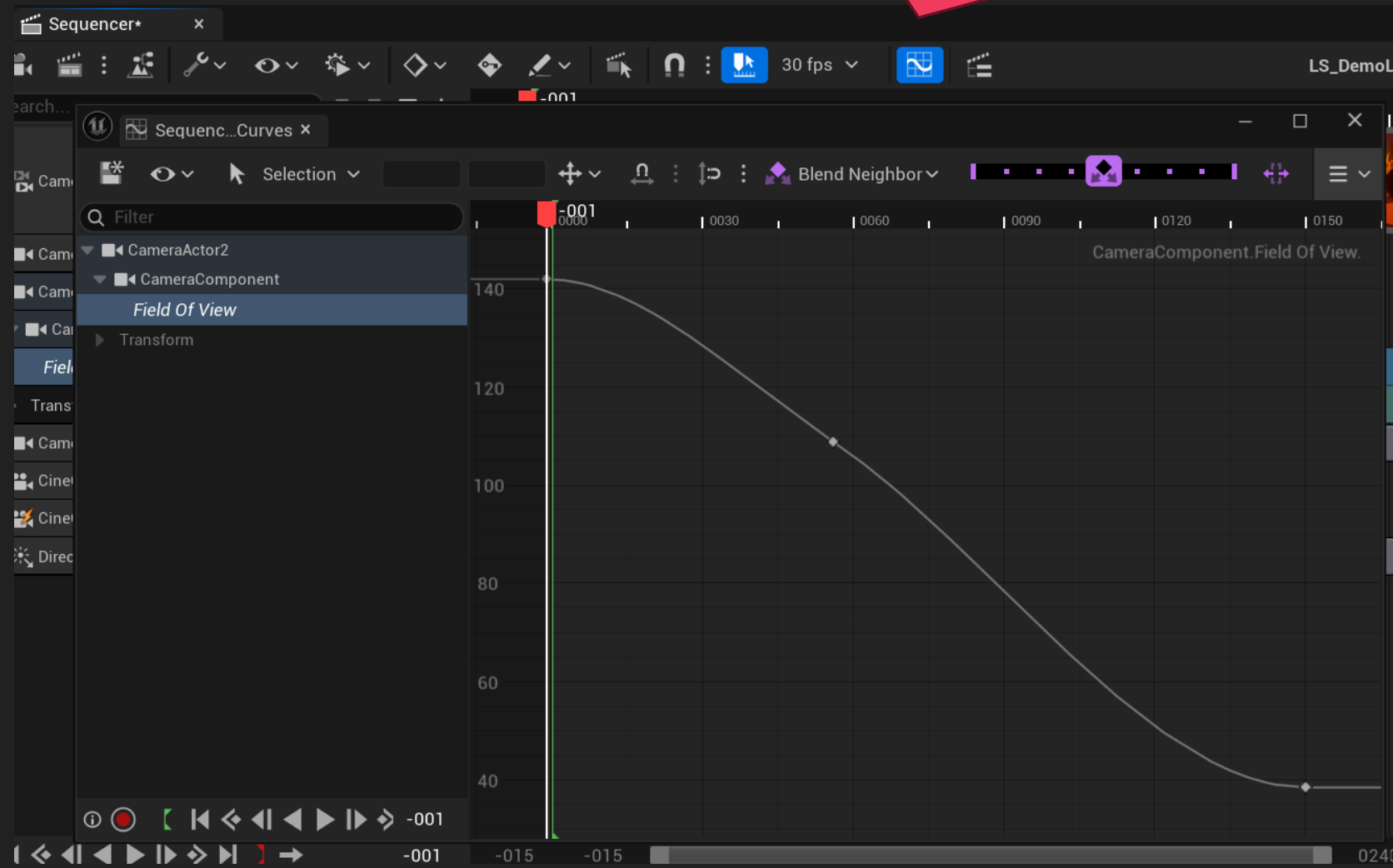
The "Snap" setting is useful for aligning clips.



Sequencer: Curves View

This icon will open up the Curves view for more control

You can open up the Curve Editor window to control HOW two keyframes transition between each other and fine tune the animation.

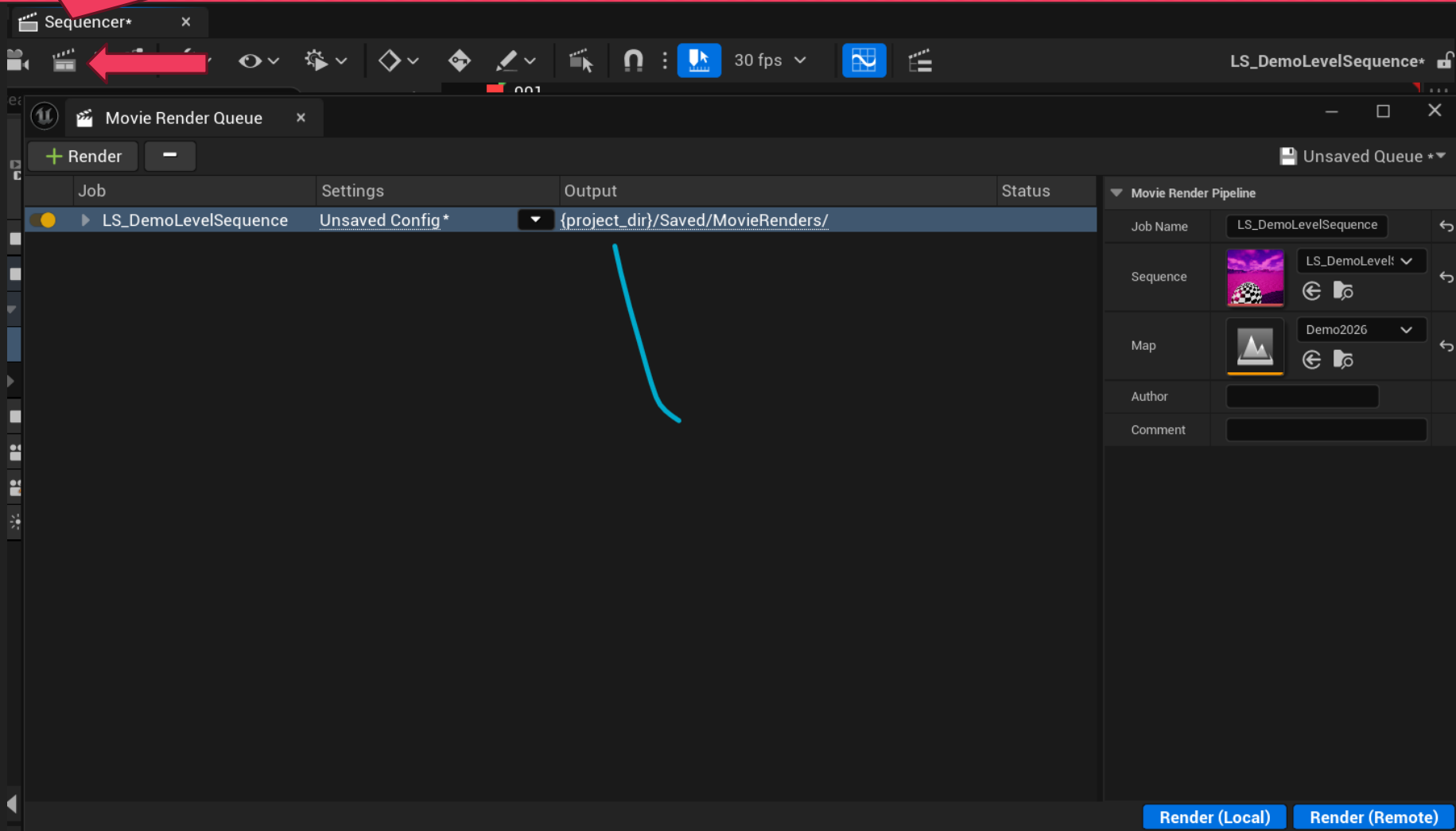


Sequencer

Open Render (new
and old options)

This icon will open the Render Queue, where you can render out your sequence as a movie file or frames.

We'll touch on this later in more detail.



Sequencer

This should be enough to get started, but this is a huge topic that could take a career to understand completely.

We will pick up with rendering next week.

Advanced Tips

+ Add

Search

					+		
					Cine Camera Actor (Lon	+	
					CameraComponent	+	
				Current A	2.8	+	
				Current F	35.0	+	
				Manual F	100000.0	+	
				Spawned	☒		
					Transform	+	
					Cine Camera Actor (Mid	+	
					Cine Camera Actor (Wid	+	
					CR_Mech	+	
					SpotLight	+	

0017

The lighting bolt means the object will be spawned by the Sequencer dynamically

-060 | 0000 | 0060 | 0120 | 0180 | 0240

Cine Camera Actor (Wide)

Each of these are a different Cine Camera Actor that will be spawned when the sequence plays.

-249* | -074*